

ON FAMILIES OF DEGENERATE REPRESENTATIONS OF $\mathrm{GL}_n(F)$

JOHANNES GIRSCH AND DAVID HELM

ABSTRACT.

1. INTRODUCTION

2. THE HIGHEST DERIVATIVE PARTITION

Let F be a finite extension of $\mathbb{Q}_p$, and let G_n denote the group $\mathrm{GL}_n(F)$. We will be interested in representations of G_n over an algebraically closed field k of finite characteristic $\ell \neq p$, over the Witt vectors $W(k)$, which we denote by $\mathcal{O}$, and over the field of fractions $\mathcal{K}$ of $\mathcal{O}$. We denote by $\mathrm{Rep}_{\mathcal{O}}(G_n)$ the category of smooth $\mathcal{O}[G_n]$ -modules, and let $\mathrm{Rep}_k(G_n)$ and $\mathrm{Rep}_{\mathcal{K}}(G_n)$ be the full subcategories of smooth $k[G_n]$ and $\mathcal{K}[G_n]$ -modules, respectively.

For $1 \leq k \leq n$ we denote by $V \mapsto V^{(k)}$ the k th derivative functor of Bernstein-Zelevinsky; it is an exact functor from $\mathrm{Rep}_{\mathcal{O}}(G_n)$ to $\mathrm{Rep}_{\mathcal{O}}(G_{n-k})$ (where we take G_0 to be the trivial group when $n = k$.)

Fix a Borel subgroup B_n of G_n with unipotent radical U_n , and a generic character $\psi : U_n \rightarrow \mathcal{O}^\times$. The representation $W_n = \mathrm{c}\text{-Ind}_{U_n}^{G_n} \psi$ is independent of these choices, and is closely related to the highest derivative functor $V \mapsto V^{(n)}$. Indeed, if V is a finitely generated $\mathcal{O}[G_n]$ -module, or more generally a module such that $eV = V$ for e a finite sum of primitive idempotents in the center Z_n of $\mathrm{Rep}_{\mathcal{O}}[G_n]$, then there is a natural isomorphism $\mathrm{Hom}_{G_n}(W_n, V) \cong V^{(n)}$. (In what follows, we will refer to such V as having “finite Bernstein support”.)

This description extends to natural description of the k th derivative functor for all k . Let $P_{k,n-k}$ be the standard parabolic of G_n whose standard Levi subgroup $M_{k,n-k}$ is block diagonal with block sizes k and $n - k$. For V with finite Bernstein support we then have an isomorphism:

$$V^{(k)} \cong \mathrm{Hom}_{G_k}(W_k, r_{G_n}^{P_{k,n-k}} V)$$

where $r_{G_n}^{P_{k,n-k}}$ denotes the parabolic restriction functor from $\mathrm{Rep}_{\mathcal{O}}(G_n)$ to $\mathrm{Rep}_{\mathcal{O}}(M_{k,n-k})$ and G_k acts on this parabolic restriction by its identification with the k by k block in $M_{k,n-k}$.

From this description it follows that one can define iterated derivatives in terms of parabolic restriction is well: if $k_1 + \dots + k_r = n$, and P is the standard parabolic with block sizes $k_1, \dots, k_r$, with Levi subgroup M , then for any V with finite Bernstein support we have an isomorphism:

$$((V^{(k_1)})^{(k_2)} \dots)^{(k_r)} \cong \mathrm{Hom}_M(W_M, r_{G_n}^P V),$$

where W_M denotes the tensor product of the W_{k_i} , considered as a representation of M . In particular this means that if k'_i is any permutation of the k_i , then one has an isomorphism:

$$((V^{(k_1)})^{(k_2)} \dots)^{(k_r)} \cong ((V^{(k'_1)})^{(k'_2)} \dots)^{(k'_r)}.$$

If V is an irreducible representation of G_n , then the highest derivative of V is the largest k such that $V^{(k)}$ is nonzero; in this case $k \geq 1$ and $V^{(k)}$ is irreducible. We can thus define a finite sequence of integers $k_1, \dots, k_r$ such that k_i is the highest derivative of V_{i-1} for all i (taking $V_0 = V$), and $k_1 + \dots + k_r = n$. We will call this (provisionally) the *highest derivative sequence* of V . From the commutativity of iterated derivatives proved in the previous paragraph, we immediately obtain:

Lemma 2.1. *Let V be an irreducible representation of G_n , and $k_1, \dots, k_r$ its highest derivative sequence. Then $k_1 \geq \dots \geq k_r$; that is, the k_i are a partition of n .*

We thus refer to the sequence $\lambda = k_1, \dots, k_r$ as above as the *highest derivative partition* of V . For λ a partition we will denote by V^λ the corresponding iterated derivative $((V^{(\lambda_1)})^{(\lambda_2)} \dots)^{(\lambda_r)}$.

The highest derivative has the following useful “multiplicativity” property:

Lemma 2.2. *Let V and W be irreducible representations of G_n and G_m , respectively, and let i and j denote the highest derivatives of V and W , respectively. Then $i + j$ is the highest derivative of the parabolic induction $i_{P_{n,m}}^{G_{n+m}} V \otimes W$, and one has a natural isomorphism:*

$$(i_{P_{n,m}}^{G_{n+m}} V \otimes W)^{(i+j)} \cong i_{P_{n-i, m-j}}^{G_{n+m-i-j}} V^{(i)} \otimes W^{(j)}.$$

From this we can inductively conclude a similar “multiplicativity” for the highest derivative partition:

Corollary 2.3. *Let V and W be irreducible representations of G_n and G_m , respectively, and let λ and λ' denote their highest derivative partitions. Then $\lambda + \lambda'$ is the highest derivative partition of $i_{P_{n,m}}^{G_{n+m}} V \otimes W$, and there is a natural isomorphism:*

$$(i_{P_{n,m}}^{G_{n+m}} V \otimes W)^{\lambda+\lambda'} \cong V^\lambda \otimes W^{\lambda'}.$$

These iterated derivative functors are represented by natural projective objects in $\text{Rep}_{\mathcal{O}}(G_n)$ (at least with respect to the subcategory of objects with finite Bernstein support). More precisely, for any λ , let P be the standard parabolic with block sizes corresponding to the λ_i , and M its standard Levi subgroup. Then, if P° denotes the opposite parabolic to P , Bernstein’s second adjointness gives an isomorphism:

$$\text{Hom}_{G_n}(i_{P^\circ}^{G_n} W_M, V) \cong \text{Hom}_M(W_M, r_{G_n}^P V),$$

and we have seen that the latter is necessarily isomorphic to V^λ for any V with finite Bernstein support.

Let W'_λ denote the parabolic induction $i_{P^\circ}^{G_n} W_M$. We have:

Theorem 2.4. *The W'_λ are projective objects of $\text{Rep}_{\mathcal{O}}(G_n)$. Moreover, for any primitive idempotent e of the center Z_n , the representation eW'_λ is finitely generated.*

Proof. Let e be a primitive idempotent of Z_n , and let e_M be a primitive summand of the corresponding idempotent of the center of $\text{Rep}_{\mathcal{O}}(M)$. Then $e_M W_M$ is a finitely generated $\mathcal{O}[M]$ -module, and the parabolic induction $i_{P^\circ}^{G_n} e_M W_M$ is isomorphic to eW'_λ . Thus eW'_λ is finitely generated. It is projective because we have an isomorphism of functors:

$$\text{Hom}_{G_n}(eW'_\lambda, V) \cong (eV)^\lambda.$$

Since W'_λ is the direct sum of the eW'_λ it is projective as well. $\square$

The representation W'_λ has an alternative description as a compact induction from a closed subset. Indeed, let ψ_λ be the character of U given by

$$\psi_\lambda(U) = \sum_{i \in S_\lambda} \psi(u_{i, i+1})$$

for some nontrivial character $\psi : F \rightarrow \mathcal{O}^\times$, where the set S_λ is given by

$$S_\lambda = \{1, 2, \dots, n\} \setminus \{\lambda_r, \lambda_r + \lambda_{r-1}, \dots, \lambda_r + \dots + \lambda_1\}.$$

We then have:

Lemma 2.5. *There is a natural isomorphism $W'_\lambda \cong \text{c-Ind}_{U_n}^{G_n} \psi_\lambda$.*

Proof. Let P' be the standard parabolic subgroup of G_n conjugate to P_λ° ; it is block upper triangular with block sizes $\lambda_r, \lambda_{r-1}, \dots, \lambda_1$. Let M' be the standard Levi subgroup of P' and $W_{M'}$ the space of Whittaker functions on M' . Then W'_λ is naturally isomorphic to $i_{P'}^{G_n} W_{M'}$.

On the other hand, we can write $W_{M'} = \text{c-Ind}_{U_n \cap M'}^{M'} \psi_\lambda|_{U_n \cap M'}$. Then $i_{P'}^{G_n} W_{M'}$ is, by definition, the representation:

$$\text{c-Ind}_{P'}^{G_n} \inf_{M'}^{P'} \text{c-Ind}_{U_n \cap M'}^{M'} \psi_\lambda|_{U_n \cap M'},$$

where $\inf$ denotes inflation from M' to P' .

But we have a natural isomorphism

$$\inf_{M'}^{P'} \text{c-Ind}_{U_n \cap M'}^{M'} \psi_\lambda|_{U_n \cap M'} \cong \text{c-Ind}_{U_n}^{P'} \psi_\lambda$$

and so the claim follows. $\square$

3. THE ZELEVINSKY CLASSIFICATION

In this section we recall the Zelevinsky classification of irreducible smooth representations of G_n . We follow the approach of Mínguez-Sécherre [MS], whose main results we now recall.

Let ν denote the cyclotomic character of $F^\times$, and let ρ be an irreducible cuspidal representation of G_n over k or $\bar{K}$. A *segment* $[a, b]_\rho$, for $b > a$ integers is a sequence of the form $[\nu^a \rho, \nu^{a+1} \rho, \dots, \nu^b \rho]$. The segments $[a, b]_\rho$ and $[a', b']_{\rho'}$ are considered equivalent if $b - a = b' - a'$ and $\nu^a \rho$ is isomorphic to $\nu^{a'} \rho'$. For a segment $\Delta = [a, b]_\rho$, where ρ is a cuspidal representation of G_n , we let $n(\Delta) = n$, $\ell(\Delta) = b - a + 1$, and $d(\Delta) = n(\Delta)\ell(\Delta)$. The *support* of Δ is the multiset $\{\nu^a \rho, \dots, \nu^b \rho\}$ of representations appearing in Δ ; note that over k this can have multiplicities above 1 if $b - a$ is at least the order of $\nu \bmod \ell$. The segment Δ is called *supercuspidal* if ρ is a supercuspidal representation of G_n .

To each segment $\Delta = [a, b]_\rho$ over k (resp. $\bar{K}$) is attached an irreducible smooth representation $Z(\Delta)$ of $G_{d(\Delta)}$ over k (resp. $\bar{K}$). If Δ is supercuspidal then the supercuspidal support of $Z(\Delta)$ is given by the support of Δ . Among the representations with this supercuspidal support, $Z(\Delta)$ is characterized by the fact that it has exactly one nonzero derivative, of degree $n(\Delta)$, and that $Z(\Delta)^{(n(\Delta))}$ is isomorphic to $Z(\Delta^-)$, where Δ^- is the segment $[a, b-1]_\rho$. (If Δ^- is empty we let $Z(\Delta^-)$ denote the one-dimensional vector space over k (resp. $\bar{K}$), with the trivial action of the trivial group G_0 .)

A *supercuspidal multisegment* over k (resp. $\bar{K}$) is a multiset of supercuspidal segments over k (resp. $\bar{K}$). To a multisegment $\mathbf{m} = \{\Delta_1, \dots, \Delta_r\}$ we associate the integer $d(\mathbf{m}) = \sum_{j=1}^r d(\Delta_j)$, and the partition $\lambda(\mathbf{m})$ of $d(\mathbf{m})$, defined by setting $\lambda(\mathbf{m})_i = \sum_{j: \ell(\Delta_j) \geq i} n(\Delta_j)$.

Let $\mathbf{m} = \{\Delta_1, \dots, \Delta_r\}$ be a supercuspidal multisegment, and let $I(\mathbf{m})$ denote the parabolic induction

$$i_P^{G_{d(\mathbf{m})}} Z(\Delta_1) \otimes Z(\Delta_2) \otimes \dots \otimes Z(\Delta_r).$$

It will be useful to us to understand the iterated derivatives of this parabolic induction. To do this we first recall the standard partial order $\preceq$ on partitions: if λ and λ' are partitions of n , we say that $\lambda \preceq \lambda'$ if for all i ,

$$\lambda_1 + \dots + \lambda_i \leq \lambda'_1 + \dots + \lambda'_i.$$

We have:

Proposition 3.1. *Let λ be a partition of $d(\mathbf{m})$. Then:*

- (1) $\text{Hom}_{G_{d(\mathbf{m})}}(W'_\lambda, I(\mathbf{m})) = 0$ unless $\lambda \preceq \lambda(\mathbf{m})$, and
- (2) $\text{Hom}_{G_{d(\mathbf{m})}}(W'_{\lambda(\mathbf{m})}, I(\mathbf{m}))$ is one-dimensional.

Proof. This is essentially [MS], Proposition 9.19. To translate their result into the language used in this paper, one must note that Mínguez-Sécherre work in terms of “residually generic” irreducible representations, rather than generic, but the two notions are equivalent for general linear groups. Thus the set they call $\text{Part}(\mathbf{m})$ is precisely the set of λ such that $r_{G_{d(\mathbf{m})}}^{P_\lambda} I(\mathbf{m})$ contains a generic representation, and by second adjointness this is the same as the set of λ such that $\text{Hom}_{G_{d(\mathbf{m})}}(W'_\lambda, I(\mathbf{m}))$ is nonzero. $\square$

From this one immediately sees:

Corollary 3.2. (1) *The parabolic induction $I(\mathbf{m})$ admits a unique irreducible subquotient $Z(\mathbf{m})$ whose highest derivative partition is equal to $\lambda(\mathbf{m})$.*

- (2) If λ is the highest derivative partition of an irreducible subquotient π of $I(\mathfrak{m})$, then $\lambda \preceq \lambda(\mathfrak{m})$.
- (3) $\text{Hom}_{G_{d(\mathfrak{m})}}(W'_\lambda, Z(\mathfrak{m})) = 0$ unless $\lambda \preceq \lambda(\mathfrak{m})$.
- (4) $\text{Hom}_{G_{d(\mathfrak{m})}}(W'_{\lambda(\mathfrak{m})}, Z(\mathfrak{m}))$ is one-dimensional.

Moreover, one has:

Theorem 3.3 ([MS], Théorème 9.36). *The map $\mathfrak{m} \mapsto Z(\mathfrak{m})$ gives a bijection between equivalence classes of supercuspidal multisegments $\mathfrak{m}$ with $d(\mathfrak{m}) = n$ over k (resp. $\bar{K}$) and irreducible smooth representations of G_n over k (resp. $\bar{K}$).*

In particular, one can conclude:

Corollary 3.4. *Let π be an irreducible representation of G_n , over an algebraically closed field of characteristic ℓ or zero, and λ its highest derivative partition. Then:*

- (1) $\text{Hom}_{G_{d(\mathfrak{m})}}(W'_\lambda, \pi) = 0$ unless $\lambda \preceq \lambda(\mathfrak{m})$.
- (2) $\text{Hom}_{G_{d(\mathfrak{m})}}(W'_{\lambda(\mathfrak{m})}, \pi)$ is one-dimensional.

4. DEGENERATE WHITTAKER MODELS

The highest derivative partition is closely related to theory of *degenerate Whittaker models*, and also to the notion of the *wave front set*. The relationship between these objects was first studied by Mœglin-Waldspurger [MW]. To any pair (S, ϕ) , where ϕ is a nilpotent element of $\text{Lie}(G_n) = M_n(F)$ and S is a semisimple element of $\text{Lie}(G_n)$ such that $[S, \phi] = -2\phi$, Mœglin-Waldspurger associate a unipotent subgroup $U_{S, \phi}$ of G_n , and a character $\psi_{S, \phi}$ of $U_{S, \phi}$ (that depends on our fixed choice of ψ above). Let $W_{S, \phi}$ be the smooth $\mathcal{O}[G_n]$ -module $\text{c-Ind}_{U_{S, \phi}}^{G_n} \psi_{S, \phi}$ (where we regard $\psi_{S, \phi}$ as a free $\mathcal{O}$ -module of rank one). We refer the reader to [GGS], particularly section 2.5, for the details of the construction; note that they work with coefficients in $\mathbb{C}$ rather than $\mathcal{O}$. The notation used here is that of [GGS] and not that of [MW].

Up to isomorphism, the space $W_{S, \phi}$ depends only on the conjugacy class of ϕ , and not on the choices of S or ψ . Since the conjugacy classes of nilpotent orbits are in bijection with partitions of n , via the Jordan decomposition, for any partition λ of n we will let W_λ denote a representation of the form $W_{S, \phi}$ for ϕ in the conjugacy class associated to λ . We call W_λ the “space of degenerate Whittaker functions” attached to λ .

A natural question to ask is, given an irreducible smooth representation π of G_n (over k or $\bar{K}$), for which λ do there exist nontrivial maps $W_\lambda \rightarrow \pi$? Over fields of characteristic zero, this question has an answer in terms of the *classification partition* of π :

Definition 4.1. Let π be an irreducible smooth representation of G_n over $\bar{K}$, let π^t denote the representation obtained by applying the Zelevinsky involution to π , and let (ϕ, N) be the Weil-Deligne representation associated to π^t via the local Langlands correspondence. (In particular, N is a nilpotent element of $\text{GL}_n(\bar{K})$.) The *classification partition* of π is the partition that is conjugate to the partition associated to the conjugacy class of N .

In the process of the proof of [GS], Theorem B, Gourevitch-Sahi show:

Proposition 4.2. *Let π be an irreducible smooth $\bar{K}$ -representation of G_n . Then the classification partition of π is equal to its highest derivative partition.*

The proof is essentially an application of the Zelevinsky classification and the additivity of the highest derivative partition with respect to parabolic induction. We refer the reader to section 6 of [GS] for the details.

The following is Theorem B of [GS]; rephrased in terms of highest derivative partitions; it is also implicit in section II.2 of [MW]:

Theorem 4.3. *Let π be an irreducible smooth representation of G_n over $\bar{K}$ and let λ be its highest derivative partition. Then:*

- (1) If λ' is any partition such that $\mathrm{Hom}_{G_n}(W_{\lambda'}, \pi)$ is nonzero, then $\lambda' \preceq \lambda$.
- (2) $\mathrm{Hom}_{G_n}(W_{\lambda}, \pi)$ is one-dimensional.

In fact, this result continues to hold in characteristic ℓ . More precisely, we have:

Theorem 4.4. *Let π be an irreducible smooth representation of G_n over k , with highest derivative partition λ . Then:*

- (1) $\mathrm{Hom}_{G_n}(W_{\lambda}, \pi)$ is one-dimensional over k , and
- (2) if λ' is any partition such that $\mathrm{Hom}_{G_n}(W_{\lambda'}, \pi)$ is nonzero, then $\lambda' \preceq \lambda$.

We will prove this theorem by lifting from characteristic ℓ to characteristic zero and applying Theorem 4.3. Let $\tilde{\pi}$ be an irreducible smooth representation of G_n over $\bar{K}$, and let $\mathcal{K}'$ be a finite extension of $\mathcal{K}$ such that $\tilde{\pi}$ is defined over $\mathcal{K}'$. Let π be a model of $\tilde{\pi}$ over $\mathcal{K}'$; that is, a smooth $\mathcal{K}'[G_n]$ -module such that $\pi \otimes_{\mathcal{K}'} \bar{K}$ is isomorphic to $\tilde{\pi}$. Let $\mathcal{O}'$ be the integral closure of $\mathcal{O}$ in $\mathcal{K}'$. We will say that $\tilde{\pi}$ is *integral* if there exists a G_n -stable $\mathcal{O}'$ -lattice L in π (this condition is independent of the choices of $\mathcal{K}'$ and the model π). If this is the case, we define $r_{\ell}\pi := L \otimes_{\mathcal{O}'} k$. The $k[G_n]$ -module $r_{\ell}\pi$ depends on the choice of lattice L , but its semisimplification is independent of these choices.

We first show:

Lemma 4.5. *For any exact sequence:*

$$0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$$

of admissible $\mathcal{O}[G_n]$ -modules, the sequence:

$$0 \rightarrow \mathrm{Hom}_{G_n}(W_{\lambda}, U) \rightarrow \mathrm{Hom}_{G_n}(W_{\lambda}, V) \rightarrow \mathrm{Hom}_{G_n}(W_{\lambda}, W) \rightarrow 0$$

is exact. In particular, let $\tilde{\pi}$ be an irreducible, smooth, integral representation of G_n over $\bar{K}$. Then

$$\dim_{\bar{K}} \mathrm{Hom}_{G_n}(W_{\lambda}, \pi) = \dim_k \mathrm{Hom}_{G_n}(W_{\lambda}, r_{\ell}\tilde{\pi}).$$

Proof. For any smooth $\mathcal{O}[G]$ -module V , let $V^{\vee}$ denote the smooth vectors in $\mathrm{Hom}_{\mathcal{O}}(V, \mathcal{K}/\mathcal{O})$; then $V \mapsto V^{\vee}$ is an exact contravariant functor and one has natural isomorphisms:

$$\mathrm{Hom}_{G_n}(V, W^{\vee}) \cong \mathrm{Hom}_{G_n}(W, V^{\vee})$$

for any W, V smooth $\mathcal{O}[G]$ -modules. If V is an admissible $\mathcal{O}[G]$ -module, then the natural map $V \rightarrow (V^{\vee})^{\vee}$ is an isomorphism.

Note that $W_{\lambda}^{\vee}$ is injective. Indeed, we have $W_{\lambda} = \mathrm{c}\text{-Ind}_{U_{S,\phi}}^{G_n} \psi_{S,\phi}$ for some S and ϕ as in the previous section, so $W_{\lambda}^{\vee}$ is isomorphic to $\mathrm{Ind}_{U_{S,\phi}}^{G_n} \psi_{S,\phi}^{-1}$, where $\psi_{S,\phi}^{-1}$ has underlying module $\mathcal{K}'/\mathcal{O}'$ and a $U_{S,\phi}$ action via the inverse of $\psi_{S,\phi}$. In particular $\psi_{S,\phi}^{-1}$ is an injective $\mathcal{O}[U_{S,\phi}]$ -module; since induction from a closed subgroup is a right adjoint it follows that $W_{\lambda}^{\vee}$ is injective as a $\mathcal{O}[G_n]$ -module.

Applying the exact functor $\mathrm{Hom}_{G_n}(-, W_{\lambda}^{\vee})$ to the exact sequence:

$$0 \rightarrow W^{\vee} \rightarrow V^{\vee} \rightarrow U^{\vee} \rightarrow 0$$

we obtain an exact sequence:

$$0 \rightarrow \mathrm{Hom}_{G_n}(U^{\vee}, W_{\lambda}^{\vee}) \rightarrow \mathrm{Hom}_{G_n}(V^{\vee}, W_{\lambda}^{\vee}) \rightarrow \mathrm{Hom}_{G_n}(W^{\vee}, W_{\lambda}^{\vee}) \rightarrow 0.$$

Invoking the isomorphisms

$$\mathrm{Hom}_{G_n}(U^{\vee}, W_{\lambda}^{\vee}) \cong \mathrm{Hom}_{G_n}(W_{\lambda}, (U^{\vee})^{\vee}) \cong \mathrm{Hom}_{G_n}(W_{\lambda}, U)$$

(and the analogous isomorphisms for V and W), we obtained the desired result.

The final statement follows by applying this exactness to the sequence:

$$0 \rightarrow L \rightarrow L \rightarrow r_{\ell}(\tilde{\pi}) \rightarrow 0,$$

where L is a $\mathcal{O}'$ lattice in some model π of $\tilde{\pi}$ over a suitable finite extension $\mathcal{K}'$ of $\mathcal{K}$. □

Remark 4.6. The lemma makes it look quite plausible that the spaces W_λ are actually projective, (and indeed, Gomez, Gourevitch, and Sahi have conjectured that this is the case) but this does not follow directly from the argument of the lemma and as far as we are aware the question remains open. Indeed, even though we can prove that $W_\lambda^\vee$ is an injective $\mathcal{O}'[G]$ -module, it does not follow that W_λ is projective; the most we can deduce from this injectivity is that W_λ is flat as a $\mathcal{O}[G_n]$ -module. When λ is the maximal partition (n) of n , then W_λ is a direct sum of finitely generated $\mathcal{O}[G_n]$ -modules, and in this case projectivity of W_λ follows from the fact that finitely generated flat smooth $\mathcal{O}[G_n]$ -modules are projective. In general however W_λ does not have good finiteness properties, and such an argument cannot be made to work.

To obtain information about $\mathrm{Hom}_{G_n}(W_\lambda, \pi)$ from lifts of π to characteristic zero we need to understand the behaviour of r_ℓ in terms of the Zelevinsky classification. We first recall that any supercuspidal representation ρ of G_n over k admits an integral, supercuspidal lift $\tilde{\rho}$ over $\bar{K}$ such that $r_\ell(\tilde{\rho}) = \rho$. A lift of Δ is a segment $\tilde{\Delta}$ of the form $[a, b]_{\tilde{\rho}}$ for some lift $\tilde{\rho}$ of ρ . The key result we need is part (1) of [MS], Théorème 9.39:

Theorem 4.7. *For any segment Δ over k , and any lift $\tilde{\Delta}$ of Δ , the reduction $r_\ell Z(\tilde{\Delta})$ is isomorphic to $Z(\Delta)$.*

Proof of Theorem 4.4: We prove this by induction on the highest derivative partition λ . If $\lambda = (1, \dots, 1)$ is minimal, then it follows from the Zelevinsky classification that $\pi = Z(\Delta)$ for some segment Δ . We can then take $\tilde{\pi} = Z(\tilde{\Delta})$; the claim then follows from Theorem 4.3 applied to $\tilde{\pi}$, followed by Lemma 4.5.

Now assume the claim holds for all representations with highest derivative partition below λ , and fix a representation $\pi = Z(\mathbf{m})$ with highest derivative partition λ . Write $\mathbf{m} = \{\Delta_1, \dots, \Delta_r\}$ and choose a lift $\tilde{\mathbf{m}} = \{\tilde{\Delta}_1, \dots, \tilde{\Delta}_r\}$ of $\mathbf{m}$. Note that $I(\tilde{\mathbf{m}})$ and $I(\mathbf{m})$ both have highest derivative partition λ . If π' is a subquotient of $I(\mathbf{m})$ other than π , then the highest derivative partition of π' lies strictly below λ , so by our inductive hypothesis $\mathrm{Hom}(W_{\lambda'}, \pi') = 0$ unless $\lambda' \prec \lambda$.

On the other hand, by Theorem 4.3, and Lemma 4.5, $\mathrm{Hom}_{G_n}(W_\lambda, I(\mathbf{m}))$ is one-dimensional; from this and the previous paragraph we deduce that $\mathrm{Hom}(W_\lambda, \pi)$ must be one-dimensional as well. Similarly, $\mathrm{Hom}(W_{\lambda'}, \pi)$ must be zero unless $\lambda' \prec \lambda$. $\square$

The results of this section suggest a strong connection between W_λ and the spaces W'_λ studied above. In fact, this can be made completely explicit. Indeed, one has:

Proposition 4.8. *There is a natural surjection $W_\lambda \rightarrow W'_\lambda$, that is noncanonically split.*

Proof. By [Go], Theorem A.1, there is a natural isomorphism of W_λ with $\mathrm{c}\text{-Ind}_{U_\lambda}^{G_n} \psi_\lambda|_{U_\lambda}$, where ψ_λ is the character defined in section 2 and U_λ is the subgroup of U_n denoted by N_λ in [Go].

We have a natural injective map:

$$\mathrm{c}\text{-Ind}_{U_n}^{G_n} \psi_\lambda^{-1} \hookrightarrow \mathrm{Ind}_{U_\lambda}^{G_n} \psi_\lambda^{-1}|_{U_\lambda}$$

that takes a left U_n, ψ_λ^{-1} -equivariant function f on G_n to the same function, but considered as a left $U_\lambda, \psi_\lambda^{-1}$ -equivariant function. Using the identity

$$\mathrm{Hom}_{G_n}(V, W^\vee) \cong \mathrm{Hom}_{G_n}(W, V^\vee),$$

together with the duality between $\mathrm{c}\text{-Ind}$ and Ind , we see that this gives rise to a map:

$$\mathrm{c}\text{-Ind}_{U_\lambda}^{G_n} \psi_\lambda|_{U_\lambda} \rightarrow \mathrm{Ind}_{U_n}^{G_n} \psi_\lambda.$$

This map is given explicitly as follows: it takes a compactly supported, left U_λ, ψ_λ -equivariant function f to the function given by:

$$g \mapsto \int_{U_\lambda \backslash U_n} \psi_\lambda^{-1}(u) f(ug) du.$$

In particular it takes compactly supported functions mod U_λ to compactly supported functions mod U_n , and thus yields a map:

$$\text{c-Ind}_{U_\lambda}^{G_n} \psi_\lambda|_{U_\lambda} \rightarrow \text{c-Ind}_{U_n}^{G_n} \psi_\lambda.$$

This is the desired map $W_\lambda \rightarrow W'_\lambda$; it remains to show that it is surjective. (It will then be split by projectivity of W'_λ .)

The dual of this map is the map:

$$\text{Ind}_{U_n}^{G_n} \psi_\lambda^{-1} \rightarrow \text{Ind}_{U_\lambda}^{G_n} \psi_\lambda^{-1}|_{U_\lambda}$$

that takes a function f to itself, forgetting part of the equivariance. This map is clearly injective, so the map $W_\lambda \rightarrow W'_\lambda$ is surjective as claimed. $\square$

5. THE SUBCATEGORY $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ AND ITS LOCALIZATION

For any partition λ of n , we can consider the full subcategories $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ and $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$, whose objects consist, respectively, of those objects V for which every simple subquotient has highest derivative partition λ' satisfying $\lambda' \preceq \lambda$ or $\lambda' \prec \lambda$.

Alternatively, one can regard $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ as the full subcategory of $\text{Rep}_\mathcal{O}(G_n)$ consisting of objects V such that $\text{Hom}_{G_n}(W'_{\lambda'}, V) = 0$ for all λ' such that $\lambda' \not\preceq \lambda$, and $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$ as the full subcategory of $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ whose objects V satisfy the additional condition $\text{Hom}_{G_n}(W'_\lambda, V) = 0$.

The inclusion of $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ in $\text{Rep}_\mathcal{O}(G_n)$ has a left adjoint $V \mapsto V^{\preceq \lambda}$. The natural map $V \rightarrow V^{\preceq \lambda}$ is surjective, and identifies $V^{\preceq \lambda}$ with the largest quotient of V that lies in $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$. In particular, the functor $V \mapsto V^{\preceq \lambda}$ sends finitely generated objects to finitely generated objects. Explicitly, one has:

$$V^{\preceq \lambda} = \text{coker } \bigoplus_{\lambda' \not\preceq \lambda} W'_{\lambda'} \otimes \text{Hom}_{G_n}(W'_{\lambda'}, V) \rightarrow V.$$

As a left adjoint of an exact functor, the functor $V \mapsto V^{\preceq \lambda}$ is right exact, and takes projective objects of $\text{Rep}_\mathcal{O}(G_n)$ to projective objects of $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ (although of course such objects will not typically be projective objects of $\text{Rep}_\mathcal{O}(G_n)$.) In particular, the $\mathcal{O}[G]$ -module $(W'_\lambda)^{\preceq \lambda}$ is a projective object of $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$.

Let $\text{Rep}_\mathcal{O}(G_n)^\lambda$ denote the Serre quotient of $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ by the Serre subcategory $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$. This is in effect a localization of $\text{Rep}_\mathcal{O}(G_n)^{\preceq \lambda}$ in which we identify all objects of $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$ with the zero object, or equivalently, in which we formally invert the maps $V \rightarrow 0$ for any V in $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$. We denote by $\text{Hom}_{G_n, \lambda}(V, W)$ the set of morphisms from V to W in $\text{Rep}_\mathcal{O}(G_n)^\lambda$.

Lemma 5.1. *For any object V of $\text{Rep}_\mathcal{O}(G_n)^\lambda$, there are a natural isomorphisms:*

$$\text{Hom}_{G_n}(W'_\lambda, V) \cong \text{Hom}_{G_n}((W'_\lambda)^{\preceq \lambda}, V) \cong \text{Hom}_{G_n, \lambda}((W'_\lambda)^{\preceq \lambda}, V)$$

Proof. The first isomorphism is immediate from the definition of $W \mapsto W^{\preceq \lambda}$ as left adjoint to the inclusion functor. For the second, note that $\text{Hom}_{G_n, \lambda}((W'_\lambda)^{\preceq \lambda}, V)$ is, by definition, the colimit over pairs W, V' of $\text{Hom}_{G_n}(W, V/V')$, where W is a subobject of $(W'_\lambda)^{\preceq \lambda}$ such that the quotient $(W'_\lambda)^{\preceq \lambda}/W$ lies in $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$ and V' is a subobject of V lying in $\text{Rep}_\mathcal{O}(G_n)^{\prec \lambda}$. Since every nonzero irreducible quotient of $(W'_\lambda)^{\preceq \lambda}$ has highest derivative partition λ , we must have $W = (W'_\lambda)^{\preceq \lambda}$. Moreover, since $(W'_\lambda)^{\preceq \lambda}$ is projective, and $\text{Hom}_{G_n}((W'_\lambda)^{\preceq \lambda}, V') = 0$, any element of $\text{Hom}_{G_n}((W'_\lambda)^{\preceq \lambda}, V/V')$ lifts uniquely to a map to V . Thus the colimit may be identified with $\text{Hom}_{G_n}((W'_\lambda)^{\preceq \lambda}, V)$ as claimed. $\square$

Hecke-forth denote the representation $(W'_\lambda)^{\preceq \lambda}$ by $\mathcal{P}_\lambda$. We let E_λ denote the endomorphism ring of $\mathcal{P}_\lambda$; the lemma implies it does not matter whether we take these endomorphisms in $\text{Rep}_\mathcal{O}(G_n)$ or in $\text{Rep}_\mathcal{O}(G_n)^\lambda$.

The projective $\mathcal{P}_\lambda$ is nearly a projective generator of $\text{Rep}_\mathcal{O}(G_n)^\lambda$; it is projective and admits a nonzero map to every object of this category, but fails to be finitely generated. To fix this, let e be an idempotent of the center Z_n of $\text{Rep}_\mathcal{O}(G_n)$, such that e is a finite sum of primitive idempotents. Then $e\mathcal{P}_\lambda$ is a projective generator of $e\text{Rep}_\mathcal{O}(G_n)^\lambda$. In particular the functor

$$V \mapsto \text{Hom}_{G_n, \lambda}(eW'_\lambda, V) \cong \text{Hom}_{G_n}(e\mathcal{P}_\lambda, V)$$

defines an equivalence between $\text{Rep}_{\mathcal{O}}(G_n)^\lambda$ and the category of eE_λ -modules.

It is therefore worth studying the structure of the rings eE_λ . We first note:

Proposition 5.2. *The ring eE_λ is a finitely generated $\mathcal{O}$ -algebra.*

Proof. As $e\mathcal{P}_\lambda$ is a finitely generated $\mathcal{O}[G_n]$ -module, it is admissible as an $eZ_n[G_n]$ -module. In particular its endomorphisms are finitely generated as an eZ_n -module. Since eZ_n is a finitely generated $\mathcal{O}$ -algebra the claim follows. $\square$

Let eZ_λ be the quotient of eZ_n that acts faithfully on eE_λ ; that is, the image of the natural map $eZ_n \rightarrow eE_\lambda$. We conjecture the following:

Conjecture 5.3. (1) *The ring eZ_λ is reduced and flat over $\mathcal{O}$.*

(2) *For each minimal prime η of eZ_λ , the map $eZ_\lambda \rightarrow eE_\lambda$ becomes an isomorphism after localizing at η .*

(3) *The natural map:*

$$eE_\lambda \rightarrow \prod_{\eta} (eE_\lambda)_{\eta}$$

in which η runs over the minimal primes of eZ_λ , is injective. (In particular, eE_λ is reduced, commutative, and flat over $\mathcal{O}$.)

(4) *$eE_\lambda \otimes_{\mathcal{O}} \bar{\mathcal{K}}$ is a smooth $\bar{\mathcal{K}}$ -algebra.*

We illustrate this conjecture when $n = 2$. When $\lambda = \{2\}$ these results are well known; in this case the natural map $eZ_\lambda \rightarrow eE_\lambda$ is an isomorphism for all e . When $\lambda = \{1, 1\}$, on the other hand, the full subcategory $\text{Rep}_{\mathcal{O}}(G_2)^{\preceq \lambda}$ is the full subcategory of representations on which the action of G_2 factors through the determinant. This category is thus equivalent to $\text{Rep}_{\mathcal{O}}(G_1)$, and the claims of Conjecture 5.3 are easily verified.

Over $\bar{\mathcal{K}}$ we can go much further, and in fact the rings $E_\lambda \otimes \bar{\mathcal{K}}$ can be described in a very explicit fashion.

We first need some terminology. We will say two segments $[a, b]_\rho$ and $[a', b']_{\rho'}$ are *inertially equivalent* if $b - a = b' - a'$ and ρ' is isomorphic to an unramified twist of ρ . An equivalence class of segments will be called an *inertial segment*. Similarly two multisegments $\mathbf{m}$ and $\mathbf{m}'$ are inertially equivalent if for any inertial segment $[\Delta]$, the multiplicities of the segments of $\mathbf{m}$ and $\mathbf{m}'$ in the class $[\Delta]$ are the same; we will call the inertial equivalence class of $\mathbf{m}$ an *inertial multisegment*, and denote it $[\mathbf{m}]$. If π is an irreducible representation of G_n over $\bar{\mathcal{K}}$, corresponding to a multisegment $\mathbf{m}$, then we let $[\pi]$ denote the corresponding inertial multisegment $[\mathbf{m}]$.

Note that the highest derivative partition $\lambda(\mathbf{m})$ is constant on the representatives of an inertial multisegment $[\mathbf{m}]$; we denote this invariant by $\lambda([\mathbf{m}])$. One similarly defines $d([\mathbf{m}])$.

For an inertial multisegment $[\mathbf{m}]$ we let $X_{[\mathbf{m}]}$ denote the free abelian group with generators X_{Δ_i} , where Δ_i are the inertial segments in $[\mathbf{m}]$. Let $W_{[\mathbf{m}]}$ denote the subgroup of permutations w of the segments Δ_i in $[\mathbf{m}]$ such that Δ_i is inertially equivalent to $\Delta_{w(i)}$ for all i . Then $W_{[\mathbf{m}]}$ acts on $X_{[\mathbf{m}]}$ by permuting the generators.

Normalize $[\mathbf{m}]$ by demanding that $\Delta_{w(i)} = \Delta_i$ for all i and all $w \in W_{[\mathbf{m}]}$. Let $\tilde{Z}(\Delta_i)$ denote the $\bar{\mathcal{K}}[X_{[\mathbf{m}]}][G_{d(\Delta_i)}]$ module obtained by twisting $Z(\Delta_i)$ by the character $g \mapsto X_{\Delta_i}^{v(\det g)}$ of $G_{d(\Delta_i)}$, and let $\tilde{I}([\mathbf{m}])$ denote the parabolic induction:

$$i_P^{G_{d([\mathbf{m}]})} \tilde{Z}(\Delta_1) \otimes \tilde{Z}(\Delta_2) \otimes \cdots \otimes \tilde{Z}(\Delta_r).$$

As an object of $\text{Rep}_{\bar{\mathcal{K}}}(G_{d([\mathbf{m}]})^{\preceq \lambda})$ the isomorphism class of $\tilde{I}([\mathbf{m}])$ depends on the ordering of the segments; on the other hand we will show that in the localization $\text{Rep}_{\bar{\mathcal{K}}}(G_{d([\mathbf{m}]})^\lambda$ such a parabolic induction is independent of the ordering.

To a homomorphism $f : \bar{\mathcal{K}}[X_{[\mathbf{m}]}] \rightarrow \bar{\mathcal{K}}$ we associate the multisegment $\mathbf{m}_f$ whose segments consists of $\Delta_1 \otimes (\chi_{f(X_1)} \circ \det), \dots, \Delta_r \otimes (\chi_{f(X_r)} \circ \det)$, where χ_α denotes the unramified character of $F^\times$ that takes a uniformizer to α . Note that for $w \in W_{[\mathbf{m}]}$ we have $\mathbf{m}_{f \circ w} = \mathbf{m}_f$. This defines a bijection between the following sets:

- (1) $W_{[\mathbf{m}]}$ -orbits of maps $\bar{\mathcal{K}}[X_{[\mathbf{m}]}] \rightarrow \bar{\mathcal{K}}$,
- (2) maps $\bar{\mathcal{K}}[X_{[\mathbf{m}]}]^{W_{[\mathbf{m}]}} \rightarrow \bar{\mathcal{K}}$
- (3) multisegments $\mathbf{m}$ in the inertial equivalence class $[\mathbf{m}]$, and
- (4) irreducible representations π of $G_{d([\mathbf{m}])}$ over $\bar{\mathcal{K}}$ with $[\pi] = [\mathbf{m}]$.

(More precisely, the bijection takes a map f first to its restriction to the ring of $W_{[\mathbf{m}]}$ -invariants, then to the corresponding multisegment $\mathbf{m}_f$, and finally to $Z(\mathbf{m}_f)$.)

Let λ be a partition of n . For any $[\mathbf{m}]$ with $[\mathbf{m}] = \lambda$, let $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^{[\mathbf{m}]}$ be the full subcategory of $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^\lambda$ consisting of representations V such that every simple subobject π of V satisfies $[\pi] = [\mathbf{m}]$. We then have:

Theorem 5.4. *Let λ be a partition of n , and let $E_{\lambda, \bar{\mathcal{K}}}$ denote the endomorphism ring of $\mathcal{P}_\lambda \otimes_{\mathcal{O}} \bar{\mathcal{K}}$.*

- (1) *There is a natural product decomposition:*

$$\text{Rep}_{\bar{\mathcal{K}}}(G_n)^\lambda \cong \prod_{\lambda([\mathbf{m}]) = \lambda} \text{Rep}_{\bar{\mathcal{K}}}(G_n)^{[\mathbf{m}]}$$

where the product is over equivalence classes of inertial multisegments $[\mathbf{m}]$ with $\lambda(\mathbf{m}) = \lambda$.

- (2) *The decomposition from part (1) induces a product decomposition:*

$$E_{\lambda, \bar{\mathcal{K}}} \cong \prod_{\lambda([\mathbf{m}]) = \lambda} E_{[\mathbf{m}]}$$

where for each $[\mathbf{m}]$ $E_{[\mathbf{m}]}$ is the endomorphism ring of the projection of $\mathcal{P}_\lambda$ to the direct factor $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^{[\mathbf{m}]}$.

- (3) *For each $[\mathbf{m}]$, there is a natural isomorphism:*

$$E_{[\mathbf{m}]} \cong \bar{\mathcal{K}}[X_{[\mathbf{m}]}]^{W_{[\mathbf{m}]}}.$$

(In particular the $E_{[\mathbf{m}]}$ are smooth, finite type, commutative $\bar{\mathcal{K}}$ -algebras.) This isomorphism is uniquely characterized by the property that for any map:

$$f : \bar{\mathcal{K}}[X_{[\mathbf{m}]}]^{W_{[\mathbf{m}]}] \rightarrow \bar{\mathcal{K}}$$

one has an isomorphism:

$$\mathcal{P}_\lambda \otimes_{E_{\lambda, \bar{\mathcal{K}}}} \bar{\mathcal{K}} \cong Z(\mathbf{m}_f)$$

in $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^{[\mathbf{m}]}$.

- (4) *Let $e_{\mathbf{m}}$ be the idempotent of $E_{\lambda, \bar{\mathcal{K}}}$ corresponding to the factor $E_{[\mathbf{m}]}$. There is a natural, $\bar{\mathcal{K}}[X_{[\mathbf{m}]}]$ -equivariant isomorphism in $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^\lambda$:*

$$e_{\mathbf{m}} \mathcal{P}_\lambda \otimes_{E_{[\mathbf{m}]}} \bar{\mathcal{K}}[X_{[\mathbf{m}]}] \cong \tilde{I}([\mathbf{m}]).$$

In particular, $\tilde{I}([\mathbf{m}])$ is a projective generator of $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^{[\mathbf{m}]}$ (and is independent, up to isomorphism, of the ordering of the segments used to construct it).

The proof of this theorem will occupy the next several sections, concluding in section 11.

6. THE ENDOMORPHISM RING $E_{\lambda, \bar{\mathcal{K}}}$

We now turn our attention to $E_{\lambda, \bar{\mathcal{K}}}$, which we will study by studying its formal completions at maximal ideals $\mathfrak{p}$ of the center $Z_{n, \bar{\mathcal{K}}}$ of $\text{Rep}_{\bar{\mathcal{K}}}(G_n)$. Since $\mathcal{P}_\lambda$ is an admissible $Z_{n, \bar{\mathcal{K}}}[G_n]$ -module, the formal completion of E_λ at $\mathfrak{p}$ may be identified with the endomorphism ring of $(\mathcal{P}_\lambda)_{\mathfrak{p}}$. We can regard this module as an object of the category $\text{Rep}_{\mathcal{O}}(G_n)_{\mathfrak{p}}^{\preceq \lambda}$ obtained by taking the pro-completion of the full subcategory of $\text{Rep}_{\mathcal{O}}(G_n)^{\preceq \lambda}$ whose objects are the finite-length $\mathcal{O}[G_n]$ -modules annihilated by some power of $\mathfrak{p}$.

Standard arguments show that there are finitely many simple objects of $\text{Rep}_{\mathcal{O}}(G_n)_{\mathfrak{p}}^{\preceq \lambda}$ (namely, the simple $\mathcal{O}[G_n]$ -modules of highest derivative partition $\lambda' \preceq \lambda$ and supercuspidal support given by $\mathfrak{p}$), and that each such simple object π admits a projective cover $\mathcal{P}_\pi \rightarrow \pi$, unique up to isomorphism.

Moreover, the indecomposable projective objects are precisely the $\mathcal{P}_\pi$. In particular, $(\mathcal{P}_\lambda)_\mathfrak{p}$ decomposes as the finite direct sum of the representations $\mathcal{P}_\pi$, where π is simple of highest derivative partition λ and supercuspidal support given by $\mathfrak{p}$.

Lemma 6.1. *Let $\mathfrak{p}$ be a maximal ideal of $Z_{n,\bar{\mathcal{K}}}$, and suppose that for any two distinct simple $\mathcal{O}[G_n]$ -modules π, π' of highest derivative λ and supercuspidal support given by $\mathfrak{p}$, we have that π' is not isomorphic to a subquotient of $\mathcal{P}_\pi$. Then we have an isomorphism:*

$$(E_\lambda)_\mathfrak{p} \cong \prod_{\pi} E_\pi$$

where π runs over all isomorphism classes of simple $\mathcal{O}[G_n]$ -modules of highest derivative partition λ and supercuspidal support given by $\mathfrak{p}$, and E_π is the endomorphism ring of $\mathcal{P}_\pi$.

Proof. We have that $(E_\lambda)_\mathfrak{p}$ is the endomorphism ring of $\bigoplus_\pi \mathcal{P}_\pi$, and the given condition implies that $\text{Hom}_{\mathcal{O}[G]}(\mathcal{P}_\pi, \mathcal{P}_{\pi'}) = 0$ unless π and π' are isomorphic to each other. $\square$

Lemma 6.2. *Let $\mathfrak{p}$ be a maximal ideal of $Z_{n,\bar{\mathcal{K}}}$, let $\mathcal{P}$ be a projective object of $\text{Rep}_{\mathcal{O}}(G)_{\bar{\mathcal{K}}}^{\leq \lambda}$ that is admissible as a $(Z_n)_\mathfrak{p}[G]$ -module, and let A be a commutative local $(Z_n)_\mathfrak{p}$ -algebra, with maximal ideal $\mathfrak{q}$ such that A acts on $\mathcal{P}$ (compatibly with the action of $(Z_n)_\mathfrak{p}$). Suppose that:*

- (1) *The cosocle of the quotient $\mathcal{P}/\mathfrak{q}\mathcal{P}$ is isomorphic to an absolutely irreducible representation π of G_n with highest derivative partition λ and supercuspidal support given by π .*
- (2) *The quotient $\mathcal{P}/\mathfrak{q}\mathcal{P}$ contains no subquotients isomorphic to π other than the cosocle.*

Then there is a quotient A' of A such that $\mathcal{P}$ is isomorphic to $\mathcal{P}_\pi \otimes_{(Z_n)_\mathfrak{p}} A'$ as an $A[G_n]$ -module, and the action of E_π on $\mathcal{P}_\pi \otimes_{(Z_n)_\mathfrak{p}} A'$ is via an injection: $E_\pi \rightarrow A'$ that makes A' into a free, finite rank E_π -module. In particular E_π is commutative.

Proof. Consider the map:

$$\mathcal{P}_\pi \otimes \text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P}) \rightarrow \mathcal{P} \rightarrow \mathcal{P}/\mathfrak{q}\mathcal{P}.$$

This map is equivariant for the action of A , and condition (1) implies that the composition is surjective. It follows that $\mathcal{P}$ is a finite direct sum of copies of $\mathcal{P}_\pi$, so that $\text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P})$ is a free E_π -module of finite rank. On the other hand, since $\mathcal{P}_\pi$ is projective, we have an isomorphism:

$$\text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P}) \otimes_A A/\mathfrak{q} \cong \text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P}/\mathfrak{q}\mathcal{P})$$

of finitely generated $A[G_n]$ -modules. Moreover the right hand side has dimension one over $A/\mathfrak{q}$ by conditions (1) and (2), so $\text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P})$ is cyclic as an A -module by Nakayama's lemma. Thus if A' denotes the quotient of A that acts faithfully on $\text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P})$, the action of E_π on $\text{Hom}_{G_n}(\mathcal{P}_\pi, \mathcal{P})$ is via a (necessarily injective) homomorphism from E_π to A' . $\square$

Our strategy for studying the rings E_π (and hence E_λ) is to construct representations $\mathcal{P}$ as in the above lemma for various π . For now we work over $\bar{\mathcal{K}}$. Let π be an irreducible $\bar{\mathcal{K}}[G]$ -module of highest derivative partition λ ; we have $\pi = Z(\mathfrak{m})$ for some multisegment $\mathfrak{m} = \{\Delta_1, \dots, \Delta_r\}$ over $\bar{\mathcal{K}}$. Write $\Delta_i = [a_i, b_i]_{\rho_i}$, and for each $1 \leq i \leq r$, and $a \leq j \leq b-1$, let $\tilde{\rho}_i \nu^j$ denote the “universal unramified twist” of $\rho_i \nu^j$ over $\bar{\mathcal{K}}[T_{ij}]^{\pm 1}$; that is, the representation $\rho_i \nu^j \otimes \chi_{ij}$, where $\chi_{ij} : F^\times \rightarrow \bar{\mathcal{K}}[T_{ij}^{\pm 1}]$ is the unramified character taking a uniformizer of $F^\times$ to T_{ij} . Then $\tilde{\rho}_i \nu^j$ is a projective $\bar{\mathcal{K}}[G_{n_i}]$ -module.

Let $\tilde{I}(\mathfrak{m})$ denote the parabolic induction:

$$i_Q^{G_n} \tilde{\rho}_1 \nu^{b_1-1} \chi_{1,b_1-1} \otimes \dots \otimes \tilde{\rho}_1 \nu^{a_1} \chi_{1,a_1} \otimes \dots \otimes \tilde{\rho}_r \nu^{b_r-1} \chi_{r,b_r-1} \otimes \dots \otimes \tilde{\rho}_r \nu^{a_r} \chi_{r,a_r},$$

where Q is a standard upper triangular parabolic subgroup of G_n . Moreover, we may assume that the Δ_i are ordered such that if for some $1 \leq i < j \leq r$ there is an integer β such that $\nu^{b_i+\beta} \rho_i \cong \nu^{b_j} \rho_j$ then $\beta \geq 0$. In particular this implies that Δ_i does not precede Δ_j for any $i > j$ (in the sense of [Ze], section 4.1). Then $\tilde{I}(\mathfrak{m})$ is an admissible $\bar{\mathcal{K}}[T_{ij}^{\pm 1}][G_n]$ -module, and it is projective as a $\bar{\mathcal{K}}[G_n]$ -module by second adjointness. Moreover, Z_n acts on $\tilde{I}(\mathfrak{m})$ via a homomorphism $Z_n \rightarrow \bar{\mathcal{K}}[T_{ij}^{\pm 1}]$ that makes $\bar{\mathcal{K}}[T_{ij}^{\pm 1}]$ into a finitely generated $Z_n \otimes \bar{\mathcal{K}}$ -module.

Let $\mathfrak{q}$ be the maximal ideal of $\mathcal{K}[T_{ij}^{\pm 1}]$ generated by $T_{ij} - 1$ for all i, j . Then $\mathfrak{q}$ lies over the maximal ideal $\mathfrak{p}$ of $Z_n \otimes \overline{\mathcal{K}}$ corresponding to the supercuspidal support of π . Set $\mathcal{P} = \tilde{I}(\mathfrak{m})_{\mathfrak{q}}^{\preceq \lambda}$; it is an admissible $\overline{\mathcal{K}}[G]$ -module that is a projective object of $\text{Rep}_{\overline{\mathcal{K}}}(G_n)_{\mathfrak{p}}^{\preceq \lambda}$. Let A be the quotient of $\overline{\mathcal{K}}[T_{ij}^{\pm 1}]_{\mathfrak{q}}$ that acts faithfully on $\mathcal{P}$.

In fact, the pair $(A, \mathcal{P})$ satisfies the conditions of Lemma 6.2. To see this, note that for each segment Δ_i , we have a natural surjection:

$$i_Q^{G_{n_i(b_i - a_i)}} (\nu^{b_i - 1} \rho_i \otimes \cdots \otimes \nu^{a_i} \rho_i) \rightarrow Z(\Delta_i)$$

for a suitable standard upper triangular parabolic P . Combining these for all i yields a surjection:

$$i_{Q'}^{G_n} (\nu^{b_1 - 1} \rho_1 \otimes \cdots \otimes \nu^{a_1} \rho_1 \otimes \cdots \otimes \nu^{b_r - 1} \rho_r \otimes \cdots \otimes \nu^{a_r} \rho_r) \rightarrow i_{Q''}^{G_n} Z(\Delta_1) \otimes \cdots \otimes Z(\Delta_r)$$

for suitable standard upper triangular parabolics Q', Q'' . Note that the parabolic induction

$$i_{P''}^{G_n} Z(\Delta_1) \otimes \cdots \otimes Z(\Delta_r)$$

is an object of $\text{Rep}_{\overline{\mathcal{K}}}(G_n)_{\mathfrak{p}}^{\preceq \lambda}$, and its unique irreducible subquotient of highest derivative partition λ is isomorphic to π . Since we have required that Δ_i does not precede Δ_j for $i > j$, this subquotient appears as the cosocle of the parabolic induction; in particular this parabolic induction is indecomposable.

We then have:

Theorem 6.3. *The surjection:*

$$i_{Q'}^{G_n} \nu^{b_1 - 1} \rho_1 \otimes \cdots \otimes \nu^{a_1} \rho_1 \otimes \cdots \otimes \nu^{b_r - 1} \rho_r \otimes \cdots \otimes \nu^{a_r} \rho_r \rightarrow i_{Q''}^{G_n} Z(\Delta_1) \otimes \cdots \otimes Z(\Delta_r)$$

induces an isomorphism of

$$(i_{Q'}^{G_n} \nu^{b_1 - 1} \rho_1 \otimes \cdots \otimes \nu^{a_1} \rho_1 \otimes \cdots \otimes \nu^{b_r - 1} \rho_r \otimes \cdots \otimes \nu^{a_r} \rho_r)^{\preceq \lambda}$$

with $i_{Q''}^{G_n} Z(\Delta_1) \otimes \cdots \otimes Z(\Delta_r)$.

The proof of this result is quite difficult and makes heavy use of the machinery of Bernstein-Zelevinsky; we postpone its proof to the appendix. This computation has powerful implications for the structure of projectives of type λ :

Lemma 6.4. *Let $(A, \mathcal{P})$ be the pair constructed above. Then $(A, \mathcal{P})$ satisfies the hypotheses of Lemma 6.2. Moreover, every irreducible subquotient of $\mathcal{P}_{\pi}$ of highest derivative partition λ is isomorphic to π .*

Proof. Note that, since $V \mapsto V^{\preceq \lambda}$ is right exact, the quotient $\mathcal{P}/\mathfrak{q}\mathcal{P}$ is isomorphic to:

$$(i_{Q'}^{G_n} \nu^{b_1 - 1} \rho_1 \otimes \cdots \otimes \nu^{a_1} \rho_1 \otimes \cdots \otimes \nu^{b_r - 1} \rho_r \otimes \cdots \otimes \nu^{a_r} \rho_r)^{\preceq \lambda}$$

and hence by Theorem 6.3, to $i_{Q''}^{G_n} Z(\Delta_1) \otimes \cdots \otimes Z(\Delta_r)$.

Since the latter is indecomposable with cosocle isomorphic to π , and no other subquotients of highest derivative partition λ , the hypotheses of Lemma 6.2 are verified. In particular $\mathcal{P}$ is a finite direct sum of copies of $\mathcal{P}_{\pi}$, so it remains to show that every irreducible subquotient of $\mathcal{P}$ of highest derivative partition λ is isomorphic to π . But every such subquotient is isomorphic to a subquotient of $\mathfrak{q}^r \mathcal{P} / \mathfrak{q}^{r+1} \mathcal{P}$, so this follows from the existence of a surjection:

$$\mathfrak{q}^r / \mathfrak{q}^{r+1} \otimes \mathcal{P} / \mathfrak{q}\mathcal{P} \rightarrow \mathfrak{q}^r \mathcal{P} / \mathfrak{q}^{r+1} \mathcal{P}.$$

□

Corollary 6.5. *Let $\mathfrak{p}$ be a maximal ideal of $Z_n \otimes \overline{\mathcal{K}}$ and let $\mathfrak{m}$ be a multisegment over $\overline{\mathcal{K}}$ with supercuspidal support $\mathfrak{p}$ and highest derivative partition λ . Then $(E_{\lambda})_{\mathfrak{p}}$ is commutative.*

Proof. By Lemma 6.1 and Lemma 6.4, $(E_{\lambda})_{\mathfrak{p}}$ is isomorphic to the product of the endomorphism rings of the $\mathcal{P}_{\pi}$, and these latter endomorphism rings are commutative by Lemma 6.2 and Lemma 6.4. □

7. FAMILIES OF TYPE λ

In this section we use the projectivity of $\mathcal{P}_\lambda$ to define certain families of representations of “type λ ”, generalizing the notion of a “co-Whittaker family” from [He2].

Definition 7.1. Let A be a commutative Noetherian $\mathcal{O}$ -algebra, and let V be an admissible $A[G_n]$ -module. We say that V is a *family of type λ* over A if the following conditions are satisfied:

- (1) V lies in $\text{Rep}_{\mathcal{O}}(G_n)^{\preceq \lambda}$,
- (2) the space $\text{Hom}_{A[G_n]}(\mathcal{P}_\lambda, V)$ is free of rank one over A , and
- (3) the map

$$\mathcal{P}_\lambda \otimes_{\mathcal{O}} \text{Hom}_{A[G_n]}(\mathcal{P}_\lambda, V) \rightarrow V$$

is surjective.

When λ is the trivial partition $\{n\}$, then being of “type λ ” is equivalent to being generic, and a family of type λ in the above sense is a co-Whittaker family in the sense of [He2], Definition 6.1.

Since $\mathcal{P}_\lambda$ is projective, if V is an $A[G_n]$ -module and $f : A \rightarrow B$ is a morphism of Noetherian $\mathcal{O}$ -algebras, then the natural map:

$$\text{Hom}_{A[G_n]}(\mathcal{P}_\lambda, V) \otimes_A B \rightarrow \text{Hom}_{B[G_n]}(\mathcal{P}_\lambda, V \otimes_A B)$$

is an isomorphism. In particular if V is a family of type λ over A , then $V \otimes_A B$ is a family of type λ over B .

Note that the space $\text{Hom}_{\mathcal{O}[G_n]}(\mathcal{P}_\lambda, \mathcal{P}_\lambda^{\preceq \lambda})$ is free of rank one over E_λ . Thus for any homomorphism $f : E_\lambda \rightarrow A$ (where A is commutative and Noetherian), the base change $\mathcal{P}_\lambda^{\preceq \lambda} \otimes_{E_\lambda, f} A$ is a family of type λ over A .

This allows us to construct a “universal” family of type λ , in a manner exactly analogous to the construction of the universal co-Whittaker family in [He2]. Indeed, let E_λ^{ab} be the quotient of E_λ by its commutator ideal (or equivalently the maximal commutative quotient of E_λ). Then any map from E_λ to a commutative ring A factors through E_λ^{ab} .

The previous paragraph shows that $\mathcal{P}_\lambda^{\preceq \lambda} \otimes_{E_\lambda} E_\lambda^{\text{ab}}$ is a family of type λ over E_λ^{ab} . We will denote this family by $\mathcal{P}_\lambda^{\preceq \lambda, \text{ab}}$. Moreover, this family is universal in the following sense: for any family V of type λ over A , the action of E_λ on $\text{Hom}_{A[G]}(\mathcal{P}_\lambda, V)$ induces a homomorphism $f_V : E_\lambda \rightarrow A$, and (after a choice of generator for $\text{Hom}_{A[G]}(\mathcal{P}_\lambda, V)$ as an A -module) the surjection

$$\mathcal{P}_\lambda \otimes_{\mathcal{O}} \text{Hom}_{A[G]}(\mathcal{P}_\lambda, V) \rightarrow V$$

yields a surjection:

$$\mathcal{P}_\lambda^{\preceq \lambda, \text{ab}} \otimes_{E_\lambda^{\text{ab}}} A \rightarrow V.$$

Moreover, applying the functor $\text{Hom}_G(\mathcal{P}_\lambda, -)$ to this surjection yields an isomorphism of free A -modules of rank one.

In general, if V and W are families of type λ over A , we will say that V dominates W if there is a surjection from V to W ; such a surjection necessarily induces an isomorphism of $\text{Hom}_G(\mathcal{P}, V)$ with $\text{Hom}_G(\mathcal{P}, W)$. We can make dominance into an equivalence relation by saying that V is equivalent to V' if there exists a W that dominates both V and V' . Then the previous paragraph shows that the family $\mathcal{P}_\lambda^{\preceq \lambda, \text{ab}}$ over E_λ^{ab} is universal for families of type λ up to dominance.

8. TANGENT VECTORS

We can use the theory of families developed above to describe the tangent vectors to a point of $\text{Spec } E_\lambda^{\text{ab}}$. For a field L that is a $\mathcal{O}$ -algebra, and a map $x : E_\lambda^{\text{ab}} \rightarrow L$, we can regard x as an L -point of $\text{Spec } E_\lambda^{\text{ab}} \otimes_{\mathcal{O}} L$, and then a tangent vector to this scheme at x is given by a map $\tilde{x} : E_\lambda^{\text{ab}} \rightarrow L[\epsilon]/\epsilon^2$ lifting x .

The theory of the previous section associates to such a $\tilde{x}$ the family $\mathcal{P}_\lambda^{\preceq \lambda, \text{ab}} \otimes_{E_\lambda^{\text{ab}}, \tilde{x}} L[\epsilon]/\epsilon^2$. Denote this family by $V_{\tilde{x}}$, and set $V_x = V_{\tilde{x}}/\epsilon V_{\tilde{x}}$ (which can alternatively be described as $\mathcal{P}_\lambda^{\preceq \lambda} \otimes_{E_\lambda, x} L$.) Then V_x has an absolute irreducible cosocle π_x of type λ .

We have an exact sequence:

$$0 \rightarrow \epsilon V_{\tilde{x}} \rightarrow V_{\tilde{x}} \rightarrow V_x \rightarrow 0.$$

Since multiplication by ϵ induces a surjection of V_x onto $\epsilon V_{\tilde{x}}$, the cosocle of $V_{\tilde{x}}$ is isomorphic to π_x ; let W denote the kernel of the surjection $\epsilon V_{\tilde{x}} \rightarrow \pi_x$. Then the above exact sequence induces an exact sequence:

$$0 \rightarrow \pi_x \rightarrow V_{\tilde{x}}/W \rightarrow V_x \rightarrow 0,$$

and hence gives a well-defined element of $\text{Ext}_{G_n}^1(V_x, \pi_x)$.

Conversely, given an element of $\text{Ext}_{L[G_n]}^1(V_x, \pi_x)$, we have an extension:

$$0 \rightarrow \pi_x \rightarrow V \rightarrow V_x \rightarrow 0,$$

which we can make into an $L[\epsilon]/\epsilon^2$ -module in which ϵ acts via the endomorphism of V given by $V \rightarrow V_x \rightarrow \pi_x \hookrightarrow V$. This makes V into a family of type λ over $L[\epsilon]/\epsilon^2$, and hence gives rise to a unique map $E_\lambda^{\text{ab}} \rightarrow L[\epsilon]/\epsilon^2$ lifting x . This thus gives a bijection between tangent vectors to $\text{Spec } E_\lambda^{\text{ab}} \otimes_{\mathcal{O}} L$ at x and elements of $\text{Ext}_{L[G_n]}^1(\pi_x, V_x)$.

In the particular case $L = \overline{\mathcal{K}}$ (or any extension thereof) we can go further using the results of section 6. In this setting, if the supercuspidal support of π_x is given by a multisegment $\mathbf{m}_x = \{\Delta_1, \dots, \Delta_r\}$, then we have seen that V_x is given by the parabolic induction $I(\mathbf{m}_x) := i_{Q''}^{G_n} Z(\Delta_1) \otimes \dots \otimes Z(\Delta_r)$, where Q'' is the standard parabolic given by Theorem 6.3 and the Δ_i are ordered as in the hypotheses to that theorem.

In this case we have:

Proposition 8.1. *Suppose L is an extension of $\overline{\mathcal{K}}$. Then the natural map: $\text{Ext}_{L[G_n]}^1(\pi_x, V_x) \rightarrow \text{Ext}_{L[G_n]}^1(V_x, V_x)$ is an isomorphism. In particular the tangent space to $\text{Spec } E_\lambda$ at x is isomorphic to $\text{Ext}_{L[G_n]}^1(V_x, V_x)$.*

Proof. This is essentially Lemma 7.1 in [Ch]; more precisely it follows by taking the smooth duals of the representations appearing in that lemma, and noting that $\text{Ext}_{L[G_n]}^1(V, W)$ is isomorphic to $\text{Ext}_{L[G_n]}^1(W^\vee, V^\vee)$. $\square$

This means in particular that we can regard tangent vectors to E_λ at x as first-order deformations of a specific family of type λ in the dominance class of π_x ; namely the “maximal” such family V_x over L .

9. THE SINGLE-SEGMENT CASE

We will use results of the previous section to show that, with coefficients in $\overline{\mathcal{K}}$, the spectra of the rings E_λ are smooth $\overline{\mathcal{K}}$ -algebras. We begin by focusing on a special case.

Let $\chi^{\text{un}} : G_n \rightarrow \overline{\mathcal{K}}[T^{\pm 1}]^\times$ be the “universal unramified character”; that is, the character taking g to $T^{v(\det g)}$, where v is the valuation on F .

As a representation of G_n , the character χ^{un} has highest derivative partition $(1, 1, \dots, 1)$, and in fact we have:

Proposition 9.1. *Let λ be the partition $(1, \dots, 1)$ of n . The character χ^{un} is a projective object of $\text{Rep}_{\overline{\mathcal{K}}}(G_n)^{\preceq \lambda}$. Moreover, if $\text{Rep}_{\overline{\mathcal{K}}}(G_n)_1$ is the block of $\text{Rep}_{\overline{\mathcal{K}}}(G_n)$ containing the trivial representation, we have an equivalence of categories between $e \text{Rep}_{\overline{\mathcal{K}}}(G_n)^{\preceq \lambda}_1$ and the category of $\overline{\mathcal{K}}[T^{\pm 1}]$ -modules, given by $V \mapsto \text{Hom}_G(\chi^{\text{un}}, V)$.*

Proof. We first show that the G -action on an object V of $\text{Rep}_{\overline{\mathcal{K}}}(G)^{\preceq \lambda}_1$ factors through the determinant. When $n = 1$ this is clear. In general, the fact that the first derivative of V is the highest implies that the restriction of V to the mirabolic subgroup P_n is obtained by inflating a representation of G_{n-1} with highest derivative partition $(1, \dots, 1)$ from G_{n-1} to P_n . Thus by induction the restriction of V to P_n factors through the determinant map, and therefore so does its restriction to any conjugate of P_n . Since these conjugates together generate G_n the claim follows.

Conversely, any V that factors through the determinant is an object of $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^{\preceq \lambda}$. Those V that additionally lie in $\text{Rep}_{\bar{\mathcal{K}}}(G_n)_1$ are thus precisely those representations of G that factor through the subgroup G_n^0 generated by all compact open subgroups of G . Thus $\text{Rep}_{\bar{\mathcal{K}}}(G_n)_1^{\preceq \lambda}$ is naturally identified with the category of $\bar{\mathcal{K}}[G_n/G_n^0]$ -modules; in particular it is clear that χ^{un} is a projective generator of this category and the result follows. $\square$

This result generalizes to a class of “minimal” cases: fix a segment $\Delta = [a, b]_\rho$; its highest derivative partition is of the form $(k, \dots, k)$ for some integer k . Denote this partition by λ' .

Consider the representation $Z(\Delta) \otimes \chi^{\text{un}}$ of G_n ; it is a smooth $\bar{\mathcal{K}}[T^{\pm 1}]$ -module. Let H denote the group of unramified characters ϕ of G_n such that $\rho \otimes \phi$ is isomorphic to ρ . Then (using χ^{un} to identify $\text{Spec } \bar{\mathcal{K}}[T^{\pm 1}]$ with the torus of unramified characters of G_n) we have an action of H on $\bar{\mathcal{K}}[T^{\pm 1}]$. Fixing isomorphisms of $Z(\Delta) \otimes \phi$ with $Z(\Delta)$ for all ϕ in H , we find that $Z(\Delta) \otimes \chi^{\text{un}}$ descends uniquely to a smooth $\bar{\mathcal{K}}[T^{\pm 1}]^H$ -module, which we denote by $Z(\Delta)^{\text{un}}$, and call the “universal unramified twist of $Z(\Delta)$ ”.

Now let $\text{Rep}_{\bar{\mathcal{K}}}(G_n)_\Delta$ denote the block of $\text{Rep}_{\bar{\mathcal{K}}}(G_n)$ containing $Z(\Delta)$. This block has a projective generator, obtained as the parabolic induction of the universal unramified twist of $\rho^{\otimes b-a+1}$ from a suitable Levi subgroup to G_n , and the endomorphisms of this parabolic induction may be naturally identified with an Iwahori Hecke algebra. This identification yields an equivalence between $\text{Rep}_{\bar{\mathcal{K}}}(G_n)_\Delta$ and $\text{Rep}_{\bar{\mathcal{K}}}(\text{GL}_{b-a+1}(F'))_1$ for a suitable unramified extension F' of F . Moreover, these equivalences are naturally compatible with parabolic induction and restriction, as well as the “top derivative” functors. From this it is not hard to see that these equivalences are compatible with the filtration we have constructed, in the sense that for any partition λ of $b-a+1$, they induce equivalences between $\text{Rep}_{\bar{\mathcal{K}}}(\text{GL}_{b-a+1}(F'))_1^{\preceq \lambda}$ and $\text{Rep}_{\bar{\mathcal{K}}}(G_n)_\Delta^{\preceq k\lambda}$.

As a result, we deduce:

Proposition 9.2. *The representation $Z(\Delta)^{\text{un}}$ is a projective object of $\text{Rep}_{\bar{\mathcal{K}}}(G_n)^{\preceq \lambda'}$, and the functor $V \mapsto \text{Hom}_{G_n}(Z(\Delta)^{\text{un}}, V)$ defines an equivalence of categories between $e(\Delta) \text{Rep}_{\bar{\mathcal{K}}}(G_n)$ and the category of $\bar{\mathcal{K}}[T^{\pm 1}]$ -modules.*

We will use these results to study homological properties of arbitrary G_n -representations; this will entail reducing to these “minimal” situations via functors of the form $V \mapsto V^{\preceq \lambda'}$. In order for this to be compatible with homological considerations we will need to understand the left derived functor $V \mapsto L^{\preceq \lambda'} V$ of this functor. The key result we will need is:

Proposition 9.3. *We have $H^i(L^{\preceq \lambda'} Z(\Delta)) = Z(\Delta)$ if $i = 0$ and zero otherwise.*

Proof. For convenience and to simplify notation, we reduce to the case of the principal block; this amounts to noting that the equivalence between $\text{Rep}_{\bar{\mathcal{K}}}(G_n)_\Delta$ and $\text{Rep}_{\bar{\mathcal{K}}}(\text{GL}_{b-a+1}(F))_1$ takes $Z(\Delta)$ to the trivial character is compatible with the functors $V \mapsto V^{\preceq \lambda}$. We thus henceforth assume that $Z(\Delta)$ is the trivial character of G_{b-a+1} , so that $\lambda = (1, \dots, 1)$.

The trivial representation of G_{b-a+1} has a projective resolution:

$$0 \rightarrow P_{b-a+1} \rightarrow \dots \rightarrow P_0 \rightarrow \bar{\mathcal{K}} \rightarrow 0$$

described by Bernstein in section 4.2 of Bernstein’s lecture notes [BR]. The projective P_i is the sum, over the dimension i faces F of the fundamental alcove of the Bruhat-Tits building of G_{b-a+1} , of $\text{ind}_{P_F}^{G_{b-a+1}} 1$, where P_F is the pointwise stabilizer of F . It is clear that $(\text{ind}_{P_F}^{G_{b-a+1}} 1)^{\preceq (1, \dots, 1)} = \bar{\mathcal{K}}$ for all facets F , and that the map:

$$(\text{ind}_{P_F}^{G_{b-a+1}} 1)^{\preceq (1, \dots, 1)} \rightarrow (\text{ind}_{P'_F}^{G_{b-a+1}} 1)^{\preceq (1, \dots, 1)}$$

induced by an inclusion of facets $F' \subset F$ is an isomorphism. From this the result follows. $\square$

10. SELF-EXT ALGEBRAS AND DIMENSIONS OF TANGENT SPACES

We now use the results of the previous section, together with the geometric lemma, to compute the self-Ext algebras $\text{Ext}_{\bar{\mathcal{K}}[G_n]}^\bullet(V_x, V_x)$ for x a $\bar{\mathcal{K}}$ -point of $\text{Spec } E_\lambda$. Let $\mathbf{m} = \{\Delta_1, \dots, \Delta_r\}$ be the corresponding multisegment, with the Δ_i ordered so that

$$V_x = i_P^{G_n} Z(\Delta_1) \otimes \cdots \otimes Z(\Delta_r).$$

Our main result is as follows:

Theorem 10.1. *The Ext algebra $\text{Ext}_{\bar{\mathcal{K}}[G_n]}^\bullet(V_x, V_x)$ is an exterior algebra over $\bar{\mathcal{K}}$ in r variables of cohomological degree one. In particular, $\text{Ext}_{\bar{\mathcal{K}}[G_n]}^i(V_x, V_x)$ is a $\binom{r}{i}$ -dimensional $\bar{\mathcal{K}}$ -vector space for $0 \leq i \leq r$, and is zero otherwise.*

From this and Proposition 8.1 one deduces:

Corollary 10.2. *The tangent space to $\text{Spec } E_\lambda$ at an $\bar{\mathcal{K}}$ -point x corresponding to a multisegment $\{\Delta_1, \dots, \Delta_r\}$ containing r segments is of dimension r .*

For a segment Δ appearing in $\mathbf{m}$, let n_Δ be the multiplicity of Δ in $\mathbf{m}$, and let $\mathbf{m}_\Delta$ be the multisegment consisting of n_Δ copies of Δ . Then, for a suitable ordering $\Delta_1, \dots, \Delta_s$ of the segments appearing in $\mathbf{m}$, and a suitable parabolic $Q = MU$ of G_n , we have:

$$V_x = i_Q^{G_n} Z(\mathbf{m}_{\Delta_1}) \otimes \cdots \otimes Z(\mathbf{m}_{\Delta_s}).$$

One then has:

Lemma 10.3 ([Ch], Lemma 7.3). *The natural map:*

$$\text{Ext}_{\bar{\mathcal{K}}[M]}^i(\otimes_j Z(\mathbf{m}_{\Delta_j}), \otimes_j Z(\mathbf{m}_{\Delta_j})) \rightarrow \text{Ext}_{\bar{\mathcal{K}}[G_n]}^i(V_x, V_x)$$

induced by functoriality of parabolic induction is an isomorphism for all i .

Proof. Up to taking smooth duals of the representations involved, this is [Ch], Lemma 7.3; the proof is via a straightforward application of the geometric lemma. $\square$

This allows us to reduce to the special case where $\mathbf{m} = \mathbf{m}_\Delta$ for some Δ ; that is, when $\mathbf{m}$ consists of some number of copies of Δ . Indeed, the Levi M is a product of general linear groups $G_{n_1}, \dots, G_{n_s}$. Suppose that for each factor we knew that $\text{Ext}_{\bar{\mathcal{K}}[G_{n_j}]}^\bullet(Z(\mathbf{m}_{\Delta_j}), Z(\mathbf{m}_{\Delta_j}))$ was a graded exterior algebra in n_{Δ_j} generators of degree 1. The algebra $\text{Ext}_{\bar{\mathcal{K}}[M]}^i(\otimes_j Z(\mathbf{m}_{\Delta_j}), \otimes_j Z(\mathbf{m}_{\Delta_j}))$ is the graded tensor product of such algebras, and this would be a graded exterior algebra in $n_{\Delta_1} + \cdots + n_{\Delta_s} = r$ generators of degree 1 as Theorem 10.1 claims.

It thus suffices to consider the case where $\mathbf{m}$ consists of r copies of Δ . In this case $Z(\mathbf{m}) = i_P^{G_n} Z(\Delta)^{\otimes r}$, for some parabolic $P = LN$ of G_n . We have a natural isomorphism:

$$\text{Ext}_{\bar{\mathcal{K}}[G_n]}^\bullet(Z(\mathbf{m}), Z(\mathbf{m})) \cong \text{Ext}_{\bar{\mathcal{K}}[L]}^\bullet(r_{G_n}^P i_P^{G_n} Z(\Delta)^{\otimes r}, Z(\Delta)^{\otimes r}).$$

We first show:

Lemma 10.4. *There exists an exact sequence of $\bar{\mathcal{K}}[L]$ -modules:*

$$0 \rightarrow K \rightarrow r_{G_n}^P i_P^{G_n} Z(\Delta)^{\otimes r} \rightarrow Q \rightarrow 0,$$

where Q has length $r!$, every irreducible subquotient of Q is isomorphic to $Z(\Delta)^{\otimes r}$, and every irreducible subquotient π of K satisfies $\text{Ext}_{\bar{\mathcal{K}}[L]}^i(\pi, Z(\Delta)^{\otimes r}) = 0$ for all $i \geq 0$.

In particular we have an isomorphism:

$$\text{Ext}_{\bar{\mathcal{K}}[L]}^\bullet(r_{G_n}^P i_P^{G_n} Z(\Delta)^{\otimes r}, Z(\Delta)^{\otimes r}) \cong \text{Ext}_{\bar{\mathcal{K}}[L]}^\bullet(Q, Z(\Delta)^{\otimes r}).$$

Proof. We apply the geometric lemma to $r_{G_n}^P i_P^{G_n} Z(\Delta)^{\otimes r}$. One finds that in the resulting filtration there are $r!$ successive quotients of the form $Z(\Delta)^{\otimes r}$, indexed by those elements of the Weyl group of G_n that normalize L . Moreover, for each successive quotient π in the filtration that is not of this form, one can check that $\text{Ext}_{\overline{\mathcal{K}}[L]}^i(\pi, Z(\Delta)^{\otimes r}) = 0$ by supercuspidal support considerations.

The geometric lemma filtration is indexed by double cosets $P \backslash G_n / P$, partially ordered via the closure relations. Refine this to a total order, so that we obtain a filtration

$$0 = J_0 \subseteq J_1 \subseteq \cdots \subseteq J_s = r_{G_n}^P i_P^{G_n} Z(\Delta)^{\otimes r}.$$

Suppose that for some i we have $J_{i+1}/J_i \cong Z(\Delta)^{\otimes r}$ and J_{i+2}/J_{i+1} a successive quotient π of the geometric lemma filtration not isomorphic to $Z(\Delta)^{\otimes r}$. Since $\text{Ext}^1(\pi, Z(\Delta)^{\otimes r}) = 0$ the exact sequence:

$$0 \rightarrow Z(\Delta)^{\otimes r} \rightarrow J_{i+2}/J_i \rightarrow \pi \rightarrow 0$$

splits. Choose an isomorphism $J_{i+1}/J_i \cong \pi \oplus Z(\Delta)^{\otimes r}$, and let J'_{i+1} be the preimage of the summand π in J_{i+2} . Then replacing J_{i+1} with J'_{i+1} gives a new filtration in which the i th and $i+1$ st successive quotients have exchanged places. Iterating this process eventually yields a filtration in which every successive quotient of $J_s/J_{s-r!}$ is isomorphic to $Z(\Delta)^{\otimes r}$ and every successive quotient of $J_{s-r!}$ is not. We may thus take $K = J_{s-r!}$ and the result follows. $\square$

Set $\lambda = \lambda(\Delta)$, and let $\text{Rep}_{\overline{\mathcal{K}}}(M)^{\preceq \lambda^r}$ denote the full subcategory of $\text{Rep}_{\overline{\mathcal{K}}}(M)$ consisting of representations such that every irreducible subquotient is of the form $\pi_1 \otimes \cdots \otimes \pi_r$, with each π_i an irreducible representation of $G_{\frac{n}{r}}$ whose highest derivative partition $\lambda(\pi_i)$ satisfies $\lambda(\pi_i) \preceq \lambda$. Similarly define a full subcategory $\text{Rep}_{\overline{\mathcal{K}}}(M)^{\prec \lambda^r}$ in which we additionally require that for every irreducible subquotient π as above, there exists an i with $\lambda(\pi_i) \prec \lambda$. As usual let $\text{Rep}_{\overline{\mathcal{K}}}(M)^{\lambda^r}$ denote the corresponding Serre quotient.

Let e be the primitive idempotent of $Z_{M, \overline{\mathcal{K}}}$ corresponding to the inertial class of the supercuspidal support of $Z(\Delta)^{\otimes r}$. Then $e \text{Rep}_{\overline{\mathcal{K}}}(M)^{\prec \lambda^r} = 0$, so we have an equality $e \text{Rep}_{\overline{\mathcal{K}}}(M)^{\preceq \lambda^r} = e \text{Rep}_{\overline{\mathcal{K}}}(M)^{\lambda}$. In particular, the representation $e \mathcal{P}_{\lambda^r}$ given by $e(\mathcal{P}_{\lambda}^{\otimes r})$ is a projective generator of $e \text{Rep}_{\overline{\mathcal{K}}}(M)^{\preceq \lambda}$, and the functor $\text{Hom}_M(e \mathcal{P}_{\lambda^r}, -)$ induces an equivalence of this category with the category of $e(E_{\lambda, \overline{\mathcal{K}}}^{\otimes r})$ -modules. By the results of the previous section this latter ring is isomorphic to $\overline{\mathcal{K}}[X_1^{\pm 1}, \dots, X_r^{\pm 1}]$. Denote this ring by B_r .

We have isomorphisms:

$$\text{Ext}_{\overline{\mathcal{K}}[G]}^{\bullet}(i_P^G Z(\Delta)^{\otimes r}, i_P^G Z(\Delta)^{\otimes r}) \cong \text{Ext}_{\overline{\mathcal{K}}[M]}^{\bullet}(Q, Z(\Delta)^{\otimes r}) \cong \text{Ext}_{\overline{\mathcal{K}}[M], \lambda}^{\bullet}(L^{\preceq \lambda^r} Q, Z(\Delta)^{\otimes r})$$

and applying the equivalence $\text{Hom}_M(e \mathcal{P}_{\lambda^r}, -)$ to the right hand side we find that it is isomorphic to $\text{Ext}_{B_r}^{\bullet}(M_Q, B_r/\mathfrak{q})$, where M_Q is the complex $\text{Hom}_{\overline{\mathcal{K}}[M]}(e \mathcal{P}_{\lambda^r}, L^{\preceq \lambda^r} Q)$ and $\mathfrak{q}$ is the maximal ideal $\langle X_1 - 1, \dots, X_r - 1 \rangle$ of B_r . It thus suffices to compute M_Q .

Lemma 10.5. *The natural map $L^{\preceq \lambda^r} Q \rightarrow Q$ is an isomorphism, and thus identifies M_Q with $\text{Hom}_{\overline{\mathcal{K}}[M]}(e \mathcal{P}_{\lambda^r}, Q)$.*

Proof. The representation Q has finite length, and every irreducible subquotient of Q is isomorphic to $Z(\Delta)^{\otimes r}$, and by Proposition 9.3 we have $H^i(L^{\preceq \lambda^r} Z(\Delta)^{\otimes r}) = 0$ for $i > 0$. We thus deduce inductively that $H^i(L^{\preceq \lambda^r} Q) = 0$ for $i > 0$, and the claim follows. $\square$

Thus we may regard M_Q as a B_r -module, rather than a complex. We have:

Lemma 10.6. *The module M_Q has length $r!$, and its support consists of the maximal ideal $\mathfrak{q}$. Moreover, $\text{Hom}_{B_r}(M_Q, B_r/\mathfrak{q})$ is one-dimensional; in particular M_Q is a cyclic B_r -module.*

Proof. Since the equivalence of $\text{Rep}_{\overline{\mathcal{K}}}(M)^{\lambda}$ with B_r -modules takes $Z(\Delta)^{\otimes r}$ to $B_r/\mathfrak{q}$, the first two claims follow from the fact that Q has length $r!$ and every irreducible subquotient of Q is isomorphic to $Z(\Delta)^{\otimes r}$.

Similarly, $\text{Hom}_{B_r}(M_q, B_r/\mathfrak{q})$ is isomorphic to $\text{Hom}_{\overline{\mathcal{K}}[M]}(Q, Z(\Delta)^{\otimes r})$. The latter is isomorphic to $\text{Hom}_{\overline{\mathcal{K}}[M]}(r_{G_n}^P i_P^{G_n} Z(\Delta)^{\otimes r}, Z(\Delta)^{\otimes r})$, and thus also to $\text{End}_{\overline{\mathcal{K}}[G_n]}(i_P^{G_n} Z(\Delta)^{\otimes r})$. But $i_P^{G_n} Z(\Delta)^{\otimes r}$ is irreducible so this endomorphism ring is one-dimensional. $\square$

Let A_r be the subring $B_r^{S_r}$, where the symmetric group S_r acts by permuting the X_i . The preimage of $\mathfrak{q}$ in S_r is a maximal ideal of A_r , which we denote by $\mathfrak{p}$. We have:

Lemma 10.7. *We have an isomorphism of B_r -modules:*

$$M_Q \cong B_r/\mathfrak{p}B_r.$$

Proof. Since M_Q is cyclic of length $r!$, and $B_r/\mathfrak{p}B_r$ also has length $r!$, it suffices to show that $\mathfrak{p}B_r$ annihilates M_Q . We have an isomorphism:

$$M_Q \cong \text{Hom}_{\overline{\mathcal{K}}[M]}(\mathcal{P}_{\lambda^r}, Q)$$

and the latter is a quotient of the space:

$$\text{Hom}_{\overline{\mathcal{K}}[M]}(\mathcal{P}_{\lambda^r}, r_{G_n}^P i_P^G Z(\Delta)^{\otimes r}),$$

which in turn is isomorphic to

$$\text{Hom}_{\overline{\mathcal{K}}[G_n]}(i_P^G \mathcal{P}_{\lambda^r}, i_P^G Z(\Delta)^{\otimes r}).$$

Consider the action of Z_n on $\text{Hom}_{\overline{\mathcal{K}}[G_n]}(i_P^G \mathcal{P}_{\lambda^r}, i_P^G Z(\Delta)^{\otimes r})$. On the one hand, the action on $i_P^G \mathcal{P}_{\lambda^r}$ is via a map $Z_n \rightarrow B_r$ whose image is A_r . On the other hand, the right hand side is annihilated by an ideal of Z_n whose image in A_r under the above map is equal to $\mathfrak{p}$. Since every element of $\text{Hom}_{\overline{\mathcal{K}}[G_n]}(i_P^G \mathcal{P}_{\lambda^r}, i_P^G Z(\Delta)^{\otimes r})$ is Z_n -equivariant it follows that $\mathfrak{p}$ annihilates both sides as claimed. $\square$

It is now straightforward to compute $\text{Ext}_{\overline{\mathcal{K}}[G_n]}^\bullet(i_P^{G_n} Z(\Delta)^{\otimes r}, i_P^{G_n} Z(\Delta)^{\otimes r})$. Indeed, we have seen that there is an isomorphism of the latter with $\text{Ext}_{B_r}^\bullet(M_q, B_r/\mathfrak{q})$, and using the previous lemma and the Hom-tensor adjunction we can rewrite this as $\text{Ext}_{A_r}^\bullet(A_r/\mathfrak{p}, A_r/\mathfrak{p})$. But since $\mathfrak{p}$ is a smooth dimension r point of $\text{Spec } A_r$, this latter algebra is a graded exterior algebra on r generators of degree 1.

11. PROOF OF THEOREM 5.4

We are now in a position to prove Theorem 5.4. Let λ be a partition of n .

Recall that if $[\mathfrak{m}]$ denotes an inertial multisegment with $\lambda([\mathfrak{m}]) = \lambda$, then we let $\tilde{I}([\mathfrak{m}])$ denote a representation of G_n of the form:

$$i_P^{G_{d([\mathfrak{m}]})} \tilde{Z}(\Delta_1) \otimes \tilde{Z}(\Delta_2) \otimes \cdots \otimes \tilde{Z}(\Delta_r).$$

We remind the reader that the isomorphism class of this representation depends on the choice of ordering of the Δ_i , but this will be harmless for what follows.

We have natural isomorphisms:

$$\text{Hom}_{\overline{\mathcal{K}}[G_n]}(\mathcal{P}_\lambda, \tilde{I}([\mathfrak{m}]) \cong \tilde{I}([\mathfrak{m}])^\lambda \cong \overline{\mathcal{K}}[X_{[\mathfrak{m}}]$$

where the second isomorphism follows from the multiplicativity of the derivative. The action of E_λ on the left hand side induces an action of E_λ on $\overline{\mathcal{K}}[X_{[\mathfrak{m}}]$, which is by a natural map:

$$f_{[\mathfrak{m}]} : E_\lambda \rightarrow \overline{\mathcal{K}}[X_{[\mathfrak{m}}].$$

We first show:

Lemma 11.1. *The map $f_{[\mathfrak{m}]}$ takes a $\overline{\mathcal{K}}$ -point x of $\text{Spec } \overline{\mathcal{K}}[X_{[\mathfrak{m}}]$ to the $\overline{\mathcal{K}}$ -point of $\text{Spec } E_\lambda$ giving the action of E_λ on $\text{Hom}_{G_n}(\mathcal{P}_\lambda, Z(\mathfrak{m}_x))$.*

In particular, $f_{[\mathfrak{m}]}$ factors through $\overline{\mathcal{K}}[X_{[\mathfrak{m}}]]^{W_{[\mathfrak{m}]}}$, and is independent of the choice of ordering of segments used to define $\tilde{I}([\mathfrak{m}])$.

Proof. The $W_{[\mathfrak{m}]}$ -invariance and independence of order are immediate from the first claim, since $\overline{\mathcal{K}}[X_{[\mathfrak{m}]}]$ is reduced. For the first claim, note that the composition $x \circ f_{[\mathfrak{m}]}$ is the map giving the action of E_λ on

$$\mathrm{Hom}_{\overline{\mathcal{K}}[G_n]}(\mathcal{P}_\lambda, \tilde{I}([\mathfrak{m}]) \otimes_{\overline{\mathcal{K}}[X_{[\mathfrak{m}]}], x} \overline{\mathcal{K}}),$$

and this space is isomorphic to $\mathrm{Hom}_{\overline{\mathcal{K}}[G_n]}(\mathcal{P}_\lambda, Z(\mathfrak{m}_x))$. $\square$

Let e be the primitive idempotent of $Z_{n, \overline{\mathcal{K}}}$ corresponding to the block containing $\tilde{I}([\mathfrak{m}])$, and let $S(e, \lambda)$ be the set of equivalence classes of inertial multisegments $[\mathfrak{m}']$ such that $\lambda([\mathfrak{m}']) = \lambda$ and $\tilde{I}([\mathfrak{m}'])$ is contained in the block corresponding to e . We can now show:

Theorem 11.2. *The map $f_{[\mathfrak{m}]} : E_\lambda \rightarrow \overline{\mathcal{K}}[X_{[\mathfrak{m}]}]^{W_{[\mathfrak{m}]}}$ identifies $\mathrm{Spec} \overline{\mathcal{K}}[X_{[\mathfrak{m}]}]^{W_{[\mathfrak{m}]}}$ with a connected component of $\mathrm{Spec} eE_\lambda$. The product, over $S(e, \lambda)$, of the maps $f_{[\mathfrak{m}']}$ identifies eE_λ with the product of the rings $\overline{\mathcal{K}}[X_{[\mathfrak{m}']}]^{W_{[\mathfrak{m}']}}$.*

Proof. The first claim follows from the second, so we focus on the second claim. Note that the maximal ideals of eE_λ are in bijection with the irreducible representations of G_n of type λ in the block corresponding to e , and hence also with multisegments of type λ . Every such multisegment is in the equivalence class of precisely one inertial multisegment $[\mathfrak{m}']$ in $S(e, \lambda)$, and the $\overline{\mathcal{K}}$ points of $\mathrm{Spec} \overline{\mathcal{K}}[X_{[\mathfrak{m}']}]^{W_{[\mathfrak{m}']}}$ are in bijection with the multisegments in this equivalence class. Thus the product of the maps $f_{[\mathfrak{m}']}$ induces a bijection of $\overline{\mathcal{K}}$ -points between $\mathrm{Spec} eE_\lambda$ and the disjoint union of the smooth schemes $\mathrm{Spec} \overline{\mathcal{K}}[X_{[\mathfrak{m}']}]^{W_{[\mathfrak{m}']}}$.

Let x be a point of $\mathrm{Spec} eE_\lambda$, and let $\mathfrak{m}_x = \{\Delta_{1,x}, \dots, \Delta_{r_x,x}\}$ denote the corresponding multisegment; let $[\mathfrak{m}']$ denote the inertial class of $\mathfrak{m}_x$. Then x lies in the image of the map from $\mathrm{Spec} \overline{\mathcal{K}}[X_{[\mathfrak{m}']}]^{W_{[\mathfrak{m}']}}$, and since this map is injective and its domain has dimension r_x , we find that the local dimension of $\mathrm{Spec} eE_\lambda$ at x is at least r_x . On the other hand by Corollary 10.2 the dimension of the tangent space to $\mathrm{Spec} eE_\lambda$ at x is r_x , so the local dimension is at most r_x . Thus x is a smooth point of dimension r_x . Since x was arbitrary, we find that $\mathrm{Spec} eE_\lambda$ is a smooth scheme.

Since a map of smooth $\overline{\mathcal{K}}$ -varieties that is a bijection on points is an isomorphism, the claim follows. $\square$

From this result parts (1), (2), and (3) of Theorem 5.4 are immediate. On the other hand, for any V in $e \mathrm{Rep}_{\overline{\mathcal{K}}}(G_n)^\lambda$, one has an isomorphism:

$$\mathcal{P}_\lambda \otimes \mathrm{Hom}_{G_n}(e\mathcal{P}_\lambda, V) \cong V.$$

Applying this for $V = \tilde{I}([\mathfrak{m}])$ we see that we have an isomorphism:

$$e\mathcal{P}_\lambda \otimes_{eE_\lambda} \overline{\mathcal{K}}[X_{[\mathfrak{m}]}] \cong \tilde{I}([\mathfrak{m}]).$$

Since $\overline{\mathcal{K}}[X_{[\mathfrak{m}]}]$ is a projective eE_λ -module, it follows that $\tilde{I}([\mathfrak{m}])$ is projective in $\mathrm{Rep}_{\overline{\mathcal{K}}}(G_n)^\lambda$, and this completes the proof of (4).

12. BERNSTEIN DUALITY AND SQUARE-INTEGRABLE SUPPORT

13. APPENDIX: PROOF OF THEOREM 6.3

In this appendix we will use the theory of Bernstein-Zelevinsky derivatives to prove Theorem 6.3. In particular, all representations will have coefficients in $\overline{\mathcal{K}}$. Firstly, we will recall some aspects of their theory and by using their techniques, deduce some results which will be needed later.

13.1. Prerequisites. For representations π_i of G_{n_i} , where $1 \leq i \leq r$, we define $\pi_1 \times \dots \times \pi_r$ to be the normalized parabolic induction

$$\pi_1 \times \dots \times \pi_r := i_{Q_{(n_1, \dots, n_r)}}^{G_{n_1 + \dots + n_r}}(\pi_1 \otimes \dots \otimes \pi_r),$$

where $Q_{(n_1, \dots, n_r)}$ is the standard parabolic subgroup of $G_{n_1 + \dots + n_r}$ with blocks of sizes $n_1, \dots, n_r$.

Let ρ be an irreducible cuspidal complex representation of G_k . Recall that for a nonempty segment $\Delta = [0, \alpha - 1]_\rho = [\rho, \nu\rho, \dots, \nu^{\alpha-1}\rho]$ we define the length of Δ as $l(\Delta) = \alpha k$. Moreover, we set

$$I(\Delta) := \nu^{\alpha-1}\rho \times \dots \times \nu\rho \times \rho$$

and

$$J(\Delta) := \rho \times \nu\rho \times \dots \times \nu^{\alpha-1}\rho.$$

Then $I(\Delta)$ has an unique irreducible quotient which we denote by $Z(\Delta)$. The unique irreducible subrepresentation of $J(\Delta)$ is also $Z(\Delta)$. If Δ is the empty set we set $I(\Delta)$ and $J(\Delta)$ to be the trivial representation of the trivial group $G_0 = \{1\}$. For $1 \leq \beta \leq \alpha - 1$ we define the segment $\Delta^{-, \beta}$ as $[0, \alpha - \beta - 1]_\rho$. If $\beta > \alpha - 1$ let $\Delta^{-, \beta}$ be the empty set. For ease of notation we often write Δ^- for $\Delta^{-, 1}$. Moreover, we set ${}^-\Delta$ to be the segment $[1, \alpha - 1]_\rho$.

Let P_k be the mirabolic subgroup of G_k . We often write $\pi|_{P_k}$ for the restriction of a representation π of G_k to the mirabolic subgroup P_k . When we say that a P_k -representation σ is contained in a representation π of G_k we mean that there is an injective P_k -homomorphism $\sigma \hookrightarrow \pi|_{P_k}$. Let U_n be the unipotent radical of P_n , i.e.

$$U_n = \{(u_{ij}) \in P_n \mid u_{ij} = \delta_{ij} \text{ for } j < n\}.$$

We define a character ψ of U_n by setting that $\psi((u_{ij})) = \psi(u_{n-1, n})$. We will need the following functors, originally defined by Bernstein and Zelevinsky. Let

$$\begin{aligned} \Psi^- : \text{Rep}_{\overline{K}}(P_n) &\rightarrow \text{Rep}_{\overline{K}}(G_{n-1}), \\ \pi &\mapsto \delta_{U_n}^{-1/2} \otimes \pi / \pi(U_n, 1) \\ \Phi^- : \text{Rep}_{\overline{K}}(P_n) &\rightarrow \text{Rep}_{\overline{K}}(P_{n-1}), \\ \pi &\mapsto \delta_{U_n}^{-1/2} \otimes \pi / \pi(U_n, \psi) \\ \Psi^+ : \text{Rep}_{\overline{K}}(G_{n-1}) &\rightarrow \text{Rep}_{\overline{K}}(P_n) \\ \pi &\mapsto \delta_{U_n}^{1/2} \otimes \pi \text{ (where } U_n \text{ acts trivially),} \\ \hat{\Phi}^+ : \text{Rep}_{\overline{K}}(P_{n-1}) &\rightarrow \text{Rep}_{\overline{K}}(P_n) \\ \pi &\mapsto \text{Ind}_{P_{n-1}U_n}^{P_n}(\delta_{U_n}^{1/2} \otimes \pi) \text{ (where } U_n \text{ acts via } \psi). \end{aligned}$$

Their basic properties can be found in Section 3 of [BZ]. Note that for $\pi \in \text{Rep}_{\overline{K}}(G_n)$ its r -th Bernstein-Zelevinsky derivative equals

$$\pi^{(r)} = \Psi^-(\Phi^-)^{r-1}(\pi|_{P_n}) \in \text{Rep}_{\overline{K}}(G_{n-r}).$$

The following is a special case of Proposition 4.6 in [BZ].

Lemma 13.1. *Let Δ_1, Δ_2 be two juxtaposed segments where Δ_2 precedes Δ_1 and $l(\Delta_1 \cup \Delta_2) = r$. There is a short exact sequence of smooth G_r -representations*

$$0 \rightarrow Z(\Delta_1, \Delta_2) \hookrightarrow Z(\Delta_1) \times Z(\Delta_2) \twoheadrightarrow Z(\Delta_2 \cup \Delta_1) \rightarrow 0.$$

Lemma 13.2. *Let $\Delta = [0, \alpha - 1]_\rho$ be a segment. Then*

$$Z([\nu^\alpha \rho], \Delta)^{(k)} = Z([\nu^\alpha \rho], \Delta^-).$$

Proof. By Lemma 13.1 we have that

$$\nu^\alpha \rho \times Z(\Delta) = Z([\nu^\alpha \rho], \Delta) + Z(\Delta \cup [\nu^\alpha \rho])$$

in the Grothendieck group. By taking the (k) -th derivative of this identity we obtain

$$Z([\nu^\alpha \rho], \Delta)^{(k)} + Z(\Delta \cup [\nu^\alpha \rho])^{(k)} = Z(\Delta) + \nu^\alpha \rho \times Z(\Delta^-),$$

and since Δ^- and $[\nu^\alpha \rho]$ are unlinked we have that $Z([\nu^\alpha \rho], \Delta^-) = \nu^\alpha \rho \times Z(\Delta^-)$ which proves the result. $\square$

Lemma 13.3. *Let Δ_1, Δ_2 be two segments, where $\Delta_1 = [0, \alpha - 1]_\rho$ where $\alpha \geq 2$ and $\Delta_2 = [\beta, -1]_\rho$ for some integer $\beta < -1$. Then there is an embedding*

$$Z(\Delta_1, \Delta_2) \hookrightarrow Z(\Delta_1^-, \Delta_2) \times \nu^{\alpha-1}\rho.$$

Proof. Note that there is an embedding $Z(\Delta_1) \hookrightarrow Z(\Delta_1^-) \times \nu^{\alpha-1}\rho$ and hence by Lemma 13.1 we have an embedding

$$Z(\Delta_1, \Delta_2) \hookrightarrow Z(\Delta_1^-) \times \nu^{\alpha-1}\rho \times Z(\Delta_2) \cong Z(\Delta_1^-) \times Z(\Delta_2) \times \nu^{\alpha-1}\rho.$$

Now $Z(\Delta_1^-) \times Z(\Delta_2)$ has two Jordan-Hölder factors, $Z(\Delta_1^-, \Delta_2)$ and $Z(\Delta_2 \cup \Delta_1^-)$. If $Z(\Delta_1, \Delta_2)$ would be contained in $Z(\Delta_2 \cup \Delta_1^-) \times \nu^{\alpha-1}\rho$ by taking the $(2k)$ -derivative we would obtain that

$$Z(\Delta_1^-, \Delta_2^-) \hookrightarrow Z(\Delta_2 \cup \Delta_1^{-,2}),$$

which is absurd. $\square$

Lemma 13.4. *Let $\Delta = [0, \alpha - 1]_\rho$ be a segment. Then the Jordan-Hölder constituents of*

$$\nu^{\alpha+1}\rho \times Z([\nu^\alpha\rho], \Delta)$$

are $Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta)$ and $Z([\nu^\alpha\rho, \nu^{\alpha+1}\rho], \Delta)$.

Proof. Let $\pi = \nu^{\alpha+1}\rho \times Z([\nu^\alpha\rho], \Delta)$. Note that $Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta)$ is the unique irreducible subrepresentation of $\nu^{\alpha+1}\rho \times \nu^\alpha\rho \times Z(\Delta)$ by Theorem 6.1 of [Ze]. We have embeddings

$$Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta) \hookrightarrow \pi \hookrightarrow \nu^{\alpha+1}\rho \times \nu^\alpha\rho \times Z(\Delta)$$

and hence $Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta)$ is the unique irreducible subrepresentation of π . First note that

$$Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta) \not\cong \pi.$$

This follows since we also have an embedding

$$Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta) \hookrightarrow Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho]) \times Z(\Delta)$$

and hence would obtain an embedding

$$\pi^{(2k)} \hookrightarrow (Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho]) \times Z(\Delta))^{(2k)}.$$

However, in the Grothendieck group we have that

$$\pi^{(2k)} = Z([\nu^\alpha\rho], \Delta^-) + \nu^{\alpha+1}\rho \times Z(\Delta^-)$$

and

$$(Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho]) \times Z(\Delta))^{(2k)} = Z(\Delta) + \nu^{\alpha+1}\rho \times Z(\Delta^-)$$

which yields a contradiction.

Now let ω be a Jordan-Hölder constituent of π different from $Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta)$. Then ω has to have highest derivative either k , $2k$ or $3k$ since $\pi^{(i)} = 0$ for $i > 3k$. Note that

$$\pi^{(3k)} = Z([\nu^{\alpha+1}\rho], [\nu^\alpha\rho], \Delta)^{(3k)} = Z(\Delta^-),$$

which implies that ω cannot have highest derivative $3k$. If ω has highest derivative k then we have to have that $\omega = Z(\Delta \cup [\nu^\alpha\rho, \nu^{\alpha+1}\rho])$ and $\omega^{(k)} = Z(\Delta \cup [\nu^\alpha\rho])$. However, by Lemma 13.2 we have

$$\pi^{(k)} = \nu^{\alpha+1}\rho \times Z([\nu^\alpha\rho], \Delta^-) + Z([\nu^\alpha\rho], \Delta)$$

and hence ω has highest derivative $2k$. Since $\pi^{(2k)} = Z([\nu^\alpha\rho], \Delta^-) + \nu^{\alpha+1}\rho \times Z(\Delta^-)$ this implies that $\omega = Z([\nu^\alpha\rho, \nu^{\alpha+1}\rho], \Delta)$. $\square$

We will need the following variants of Propositions 5.1 and 5.3 of [Ze]. We refer to Section 3 of [Ze] for the necessary definitions.

Lemma 13.5 ([Ze], 5.1). *Let τ be a smooth representation of P_n . The following are equivalent:*

- (1) *All nonzero subrepresentations of τ have highest derivative at least m .*
- (2) *There is an injective P_n -morphism $\tau \rightarrow \bigoplus_{j \geq m} (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)})$.*

Proof. Note that for any smooth representations σ and ω of P_n we have by Section 3.2 of [BZ] that

$$(13.1) \quad \text{Hom}_{P_n}(\sigma, (\hat{\Phi}^+)^{r-1} \circ \Psi^+(\omega^{(r)})) \cong \text{Hom}_{G_{n-r}}(\sigma^{(r)}, \omega^{(r)})$$

for $1 \leq r \leq n$.

(1) implies (2): For any $1 \leq r \leq n$ there is a canonical map $\tau \rightarrow (\hat{\Phi}^+)^{r-1} \Psi^+(\tau^{(r)})$ which corresponds to the identity morphism $\tau^{(r)} \rightarrow \tau^{(r)}$ under the above isomorphism. Let Ω be the kernel of the induced map $\tau \rightarrow \bigoplus_{j \geq m} (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)})$. For all $j \geq m$ the composition of $\Omega \hookrightarrow \tau$ and $\tau \rightarrow (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)})$ is zero. Equation (13.1) implies that the embedding $\Omega^{(j)} \hookrightarrow \tau^{(j)}$ is the zero map and hence $\Omega^{(j)} = 0$ for all $j \geq m$. By our assumption we obtain that $\Omega = 0$.

(2) implies (1): Let σ be a nonzero subrepresentation of τ . Then there is $r \geq m$ such that the image of σ under the map $\tau \rightarrow (\hat{\Phi}^+)^{r-1} \Psi^+(\tau^{(r)})$ is nonzero. In particular we obtain that $\text{Hom}_{P_n}(\sigma, (\hat{\Phi}^+)^{r-1} \circ \Psi^+(\tau^{(r)}))$ is nonzero, which by Equation (13.1) implies that $\sigma^{(r)} \neq 0$. $\square$

This result allows us to prove the following.

Lemma 13.6 ([Ze], 5.3). *Let τ be a smooth nonzero representation of P_n and π a smooth nonzero representation of G_k . Suppose all subrepresentations of τ have highest derivative at least m , then all subrepresentations of $\pi \times \tau$ have highest derivative at least m .*

Proof. By assumption and the above proposition we have an injective map

$$\tau \rightarrow \bigoplus_{j \geq m} (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)}).$$

By exactness of $\pi \times \dots$ we see that $\pi \times \tau \rightarrow \pi \times \bigoplus_{j \geq m} (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)})$ is injective and note that

$$\pi \times \bigoplus_{j \geq m} (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)}) \cong \bigoplus_{j \geq m} \pi \times (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)}).$$

As in 5.3 of [Ze] we have an embedding $\pi \times (\hat{\Phi}^+)^{j-1} \Psi^+(\tau^{(j)}) \hookrightarrow (\hat{\Phi}^+)^{j-1} \Psi^+((\pi \times \tau)^{(j)})$ for all $j \geq m$ and hence obtain an injective map

$$\pi \times \tau \rightarrow \bigoplus_{j \geq m} (\hat{\Phi}^+)^{j-1} \Psi^+((\pi \times \tau)^{(j)}),$$

which by the above proposition implies the result. $\square$

Lemma 13.7. *Let π (and σ) be a smooth representation of G_n (respectively G_m).*

- (1) *If ω is a nonzero submodule of $(\sigma \times \pi)|_{P_{m+n}}$, then there is $r \geq 0$ such that $(\Phi^-)^r(\omega) \neq 0$ and an embedding $(\Phi^-)^r(\omega) \hookrightarrow \sigma^{(r)} \times \pi|_{P_n}$.*
- (2) *Suppose that all nonzero subrepresentations of $\pi|_{P_n}$ have highest derivative at least μ . Then all nonzero subrepresentations of $(\sigma \times \pi)|_{P_{m+n}}$ have highest derivative at least μ .*

Proof. ad (1): Assume the contrary. By Proposition 4.13a) of [BZ] there is a short exact sequence

$$0 \rightarrow \sigma|_{P_m} \times \pi \rightarrow (\sigma \times \pi)|_{P_{m+n}} \rightarrow \sigma \times (\pi|_{P_n}) \rightarrow 0,$$

and by our assumption ω cannot be contained in $\sigma^{(0)} \times (\pi|_{P_n}) = \sigma \times (\pi|_{P_n})$. Hence we have an embedding $\omega \hookrightarrow \sigma|_{P_m} \times \pi$. For $i \geq 0$ suppose that $(\Phi^-)^i(\omega) \neq 0$ and $(\Phi^-)^i(\omega)$ can be embedded into $(\Phi^-)^i(\sigma|_{P_m}) \times \pi$. By Proposition 4.13 d) of [BZ] we obtain that $(\Phi^-)^{i+1}(\omega) \neq 0$. By Proposition 4.13c) of [BZ] we have a short exact sequence

$$0 \rightarrow (\Phi^-)^{i+1}(\sigma|_{P_m}) \times \pi \rightarrow \Phi^-((\Phi^-)^i(\sigma|_{P_m}) \times \pi) \rightarrow \sigma^{(i+1)} \times \pi|_{P_n} \rightarrow 0,$$

and hence $(\Phi^-)^{i+1}(\omega)$ is contained in either $(\Phi^-)^{i+1}(\sigma|_{P_m}) \times \pi$ or in $\sigma^{(i+1)} \times \pi|_{P_n}$. However, by our assumption the latter case is impossible and hence there is an embedding $(\Phi^-)^{i+1}(\omega) \hookrightarrow (\Phi^-)^{i+1}(\sigma|_{P_m}) \times \pi$. Inductively, we obtain that $(\Phi^-)^{m-1}(\omega) \neq 0$ and an embedding $(\Phi^-)^{m-1}(\omega) \hookrightarrow (\Phi^-)^{m-1}(\sigma|_{P_m}) \times \pi$. Again by Proposition 4.13 d) [BZ] we see that $(\Phi^-)^m(\omega) \neq 0$ and obtain an embedding

$$(\Phi^-)^m(\omega) \hookrightarrow \Phi^-((\Phi^-)^{m-1}(\sigma|_{P_m}) \times \pi).$$

However, $\Phi^-((\Phi^-)^{m-1}(\sigma|_{P_m}) \times \pi)$ is isomorphic to $\sigma^{(m)} \times \pi|_{P_n}$ which yields a contradiction.

ad (2): Let ω' be a nonzero subrepresentations of $(\sigma \times \pi)|_{P_{n+m}}$. By (1) there is an integer $r \geq 0$ such that $(\Phi^-)^r(\omega') \neq 0$ and there is an embedding $(\Phi^-)^r(\omega') \hookrightarrow \sigma^{(r)} \times \pi|_{P_n}$. By Lemma 13.6 every nonzero subrepresentation of $\sigma^{(r)} \times \pi|_{P_n}$ has highest derivative at least μ . This implies that ω' has highest derivative at least $\mu + r$, which finishes the proof. $\square$

Lemma 13.8. *Let π be a smooth finite length representation of G_n and suppose that all nonzero subrepresentations of $\pi|_{P_n}$ have highest derivative at least μ . Let Δ be the segment $[\rho, \nu\rho, \dots, \nu^{\alpha-1}\rho]$ and suppose that $\nu^\alpha\rho$ is not an element of the supercuspidal support of any irreducible subquotient of π .*

- (1) *For any permutation $s \in S_\alpha$ we set Σ_s to be the representation $\Sigma_s = \nu^{s(1)-1}\rho \times \nu^{s(2)-1}\rho \times \dots \times \nu^{s(\alpha)-1}\rho$. Then all nonzero subrepresentations of $(\Sigma_s \times \pi)|_{P_{\alpha k+n}}$ have highest derivative at least $k + \mu$.*
- (2) *Let σ be an irreducible representation of $G_{\alpha k}$ with supercuspidal support Δ . Then all nonzero subrepresentations of $(\sigma \times \pi)|_{P_{\alpha k+n}}$ have highest derivative at least $k + \mu$.*

Proof. ad (1): Let ω be an irreducible subrepresentation of $(\Sigma_s \times \pi)|_{P_{\alpha k+n}}$ with highest derivative l . By Lemma 13.7 there is an integer $r \geq 0$ such that $(\Phi^-)^r(\omega) \neq 0$ and $(\Phi^-)^r(\omega) \hookrightarrow \Sigma_s^{(r)} \times \pi|_{P_n}$. Firstly, suppose that $r = 0$. Then $\omega^{(l)}$ is contained in $(\Sigma_s \times (\pi|_{P_n}))^{(l)}$ which by Corollary 4.14 of [BZ] equals $\Sigma_s \times \pi^{(l)}$. By Lemma 4.7 of [BZ] this would imply that $\nu^\alpha\rho$ is an element of the supercuspidal support of an irreducible subquotient of $\Sigma_s \times \pi$, which contradicts our assumptions.

Note that for $1 \leq j < k$ the representation $\Sigma_s^{(j)}$ is zero. Hence we obtain that $r \geq k$. By Lemma 13.6 every nonzero irreducible subrepresentation of $\Sigma_s^{(r)} \times \pi|_{P_n}$ has highest derivative at least μ so in particular $(\Phi^-)^r(\omega)$ has highest derivative at least μ . Then ω has highest derivative at least $k + \mu$.

ad (2): This follows immediately from part (1), since, by the theory of Bernstein and Zelevinsky, for any irreducible $G_{\alpha k}$ -representation σ with supercuspidal support Δ there exists a permutation $s \in S_\alpha$ such that there is an embedding of σ into Σ_s . $\square$

Lemma 13.9. *Let π be a smooth representation of G_n such that every nonzero subrepresentation of $\pi|_{P_n}$ has highest derivative at least μ . Let $\Delta = [0, \alpha - 1]_\rho$ be a segment of length at least two and suppose that $\nu^\alpha\rho$ is not an element of the supercuspidal support of any irreducible subquotient of π . Then all nonzero subrepresentations of $(Z([\nu^{\alpha-1}\rho], \Delta^-) \times \pi)|_P$ have highest derivative at least $2k + \mu$.*

Proof. Let ω be a nonzero irreducible subrepresentation of $(Z([\nu^{\alpha-1}\rho], \Delta^-) \times \pi)|_P$ with highest derivative $l \geq 1$. Lemma 13.8 yields that $l \geq k + \mu$. By Lemma 13.7 there is an integer $r \geq 0$ such that $(\Phi^-)^r(\omega) \neq 0$ and

$$(\Phi^-)^r(\omega) \hookrightarrow Z([\nu^{\alpha-1}\rho], \Delta^-)^{(r)} \times \pi|_{P_n}.$$

Note that for $r < 2k$ we have that $Z([\nu^{\alpha-1}\rho], \Delta^-)^{(r)} = 0$ unless r is 0 or k .

If $r = 0$ we see by Corollary 4.14 of [BZ] that $\omega^{(l)}$ is contained in

$$(Z([\nu^{\alpha-1}\rho], \Delta^-) \times \pi|_{P_n})^{(l)} = Z([\nu^{\alpha-1}\rho], \Delta^-) \times \pi^{(l)}.$$

By Lemma 4.7 of [BZ] this would imply that $\nu^\alpha\rho$ is an element of the supercuspidal support of an irreducible subquotient of $Z([\nu^{\alpha-1}\rho], \Delta^-) \times \pi$, which contradicts our assumptions.

If $r = k$ we have an embedding $(\Phi^-)^k(\omega) \hookrightarrow Z([\nu^{\alpha-1}\rho], \Delta^-)^{(k)} \times \pi|_{P_n}$. By Lemma 13.2 we have that

$$Z([\nu^{\alpha-1}\rho], \Delta^-)^{(k)} = Z([\nu^{\alpha-1}\rho], \Delta^{-,2})$$

and obtain an embedding

$$\omega^{(l)} \hookrightarrow Z([\nu^{\alpha-1}\rho], \Delta^{-,2}) \times \pi^{(l-k)}.$$

As before, we would have by Lemma 4.7 of [BZ] that $\nu^\alpha\rho$ is contained in the supercuspidal support of an irreducible subquotient of $Z([\nu^{\alpha-1}\rho], \Delta^-) \times \pi$, which again contradicts our assumptions.

Hence we obtain that $r \geq 2k$. By Lemma 13.6 any nonzero subrepresentation of

$$Z([\nu^{\alpha-1}\rho], \Delta^-)^{(r)} \times \pi|_{P_n}$$

has highest derivative at least μ . In particular $(\Phi^-)^r(\omega)$ has highest derivative at least μ , which immediately shows that ω has highest derivative at least $r + \mu \geq 2k + \mu$. $\square$

Lemma 13.10. *Let σ be an irreducible P_n -subrepresentation which can be embedded into a finite length G_n -representation π . Then the highest derivative sequence of σ is a partition.*

Proof. We have that σ is an irreducible representation of $\rho|_{P_n}$, where ρ is a Jordan-Hölder factor of π . However, by Corollary 6.8 of [Ze], ρ is a homogeneous representation with highest derivative $\rho^{(k)}$ for some natural number k . Then $\sigma^{(k)} \cong \rho^{(k)}$ and hence the classifying sequence of σ coincides with the classifying partition of ρ . $\square$

13.2. Technical Result. The following Proposition 13.11 is the main technical result we will need. Firstly we will set up some notation. For two cuspidal representations ρ, ρ' of G_k we say that $\rho' \geq \rho$ if either

- there is no integer β such that $\nu^\beta \rho \cong \rho'$,
- or if there is some integer β such that $\nu^\beta \rho \cong \rho'$ then $\beta \geq 0$.

Let $\Delta'_1, \dots, \Delta'_r$ be a sequence of segments such that $\Delta'_i = [\rho_i, \nu\rho_i, \dots, \nu^{\alpha_i-1}\rho_i]$, where ρ_i is a cuspidal representation of G_{n_i} . Set $n = \sum_{i=1}^r \alpha_i n_i$. We assume that $\nu^{\alpha_i-1}\rho_i \geq \nu^{\alpha_j-1}\rho_j$ for $i \leq j$. In particular this implies that Δ'_i does not precede Δ'_j if $i < j$. Let $\lambda = (\lambda_1, \lambda_2, \dots)$ be the highest derivative partition of $Z(\Delta'_1, \dots, \Delta'_r)$. Note that λ_i is the highest derivative of $Z((\Delta'_1)^{-,i-1}, \dots, (\Delta'_r)^{-,i-1})$.

For $1 \leq i \leq r$ let π_i be an irreducible representation such that the supercuspidal support of π_i equals Δ'_i . We have that

$$\pi_i = Z(\Lambda_1^i, \dots, \Lambda_{k_i}^i),$$

where Λ_j^i are disjoint segments whose union is Δ'_i and Λ_{j+1}^i precedes Λ_j^i . We also set that $\pi_i^- = Z((\Lambda_1^i)^-, \dots, (\Lambda_{k_i}^i)^-)$. Let

$$J(\pi_i) = J(\Lambda_1^i) \times J(\Lambda_2^i) \times \dots \times J(\Lambda_{k_i}^i)$$

and similarly

$$J(\pi_i^-) = J((\Lambda_1^i)^-) \times J((\Lambda_2^i)^-) \times \dots \times J((\Lambda_{k_i}^i)^-).$$

For any sequence of segments $\Delta'_1, \dots, \Delta'_r$ and representations $\pi_1, \dots, \pi_r$ with conditions as above and if there is $1 \leq s \leq r$ such that $\pi_s \not\cong Z(\Delta'_s)$ we define the representation $\mathbb{X}(\pi_1, \dots, \pi_r, s)$ as

$$\left(\prod_{t=1}^{s-1} J(\pi_t) \times \left(Z([\nu^{\alpha_s-l}\rho_s], \Lambda_2^s) \times J(-\Lambda_1^s) \times \prod_{t=3}^{k_s} J(\Lambda_t^s) \right) \times \prod_{t=s+1}^r J(\pi_t) \right) |_P.$$

Proposition 13.11. *Let ω be a nonzero irreducible subrepresentation of $(\pi_1 \times \dots \times \pi_r)|_{P_n}$. If ω has highest derivative partition smaller than or equal to λ (with respect to the lexicographic ordering), then $\pi_i = Z(\Delta'_i)$ for all $i = 1, \dots, r$.*

Proof of Proposition 13.11. Assume that not all π_i are isomorphic to $Z(\Delta'_i)$ for $1 \leq i \leq r$. By the theory of Bernstein and Zelevinsky we have a map

$$\pi_i \hookrightarrow J(\pi_i).$$

Let Γ be the set of integers $1 \leq i \leq r$ where $k_i > 1$, i.e. those integers $1 \leq i \leq r$ such that $\pi_i \not\cong Z(\Delta'_i)$. By our assumption Γ is nonempty. Choose $m \in \Gamma$ such that the length of Λ_1^m is minimal among the lengths of Λ_1^i for $i \in \Gamma$. Set l to be the length of Λ_1^m , i.e. $\Lambda_1^m = [\nu^{\alpha_m-l}\rho_m, \dots, \nu^{\alpha_m-1}\rho_m]$. By our assumption, note that l is strictly smaller than the length of Δ'_m . The theory of Bernstein-Zelevinsky yields an embedding

$$\pi_m \hookrightarrow Z(\Lambda_1^m, \Lambda_2^m) \times J(\Lambda_3^m) \times \dots \times J(\Lambda_{k_m}^m).$$

Since $-\Lambda_1^m = [\nu^{\alpha_m-l+1}\rho_m, \dots, \nu^{\alpha_m-1}\rho_m]$ and hence $\Lambda_1^m = [\nu^{\alpha_m-l}\rho_m] \cup -\Lambda_1^m$, we obtain by repeatedly applying Lemma 13.3 an embedding

$$Z(\Lambda_1^m, \Lambda_2^m) \hookrightarrow Z([\nu^{\alpha_m-l}\rho_m], \Lambda_2^m) \times J(-\Lambda_1^m).$$

Overall the above arguments yield an embedding of ω into $\mathbb{X}(\pi_1, \dots, \pi_r, m)$, i.e.

$$(13.2) \quad \left(\prod_{t=1}^{m-1} J(\pi_t) \times \left(Z([\nu^{\alpha_m-l} \rho_m], \Lambda_2^m) \times J(-\Lambda_1^m) \times \prod_{t=3}^{k_m} J(\Lambda_t^m) \right) \times \prod_{t=m+1}^r J(\pi_t) \right) |_{P_n}.$$

However, by the following Lemma 13.12 (whose proof appears at the end of this section), the highest derivative partition of ω has to be larger than λ (with respect to the lexicographic ordering). $\square$

Lemma 13.12. *Let σ be an irreducible subrepresentation of $\mathbb{X}(\pi_1, \dots, \pi_r, m)$, i.e. Equation (13.2), with highest derivative partition $\lambda^\sigma = (\lambda_1^\sigma, \lambda_2^\sigma, \dots)$. We have that $\lambda_i^\sigma \geq \lambda_i$ for all $1 \leq i \leq l$ and either*

- *there is an $1 \leq j < l$ such that $\lambda_j^\sigma > \lambda_j$,*
- *or $\lambda_l^\sigma > \lambda_l + n_m$.*

We will prove this result via induction on l , for which we need the following lemmata.

Lemma 13.13. *If $l = 1$, i.e. $\Lambda_1^m = [\nu^{\alpha_m-1} \rho_m]$, any nonzero subrepresentation σ of $\mathbb{X}(\pi_1, \dots, \pi_r, m)$ has highest derivative at least $\lambda_1 + n_m$.*

Lemma 13.14. *Suppose that $l > 1$. Then any nonzero subrepresentation σ of $\mathbb{X}(\pi_1, \dots, \pi_r, m)$ has highest derivative at least λ_1 . If the highest derivative of σ equals λ_1 , we have that*

$$\sigma^{(\lambda_1)} \hookrightarrow \mathbb{X}(\pi_1^-, \dots, \pi_r^-, m).$$

To prove Lemma 13.13 and Lemma 13.14 we introduce some new notation. For integers $1 \leq \beta \leq r$ and $1 \leq i \leq j$ if $j \leq k_\beta$ let $Y_{i,j}^\beta$ be the representation $\prod_{t=i}^j J(\Lambda_t^\beta)$ and if $j > k_\beta$ we set $Y_{i,j}^\beta$ to be the trivial representation of $G_0 = \{1\}$. Moreover, for integers $1 \leq i \leq j \leq m$ we define

$$X_{i,j} := \prod_{t=i}^j J(\pi_t)$$

if $m < i$ or $m > j$ and

$$X_{i,j} := \prod_{t=i}^{m-1} J(\pi_t) \times (Z([\nu^{\alpha_m-l} \rho_m], \Lambda_2^m) \times J(-\Lambda_1^m) \times Y_{3,k_m}^m) \times \prod_{t=m+1}^j J(\pi_t)$$

if $i \leq m \leq j$. If $l > 1$ we set

$$X_{i,j}^- := \prod_{t=i}^j J(\pi_t^-)$$

if $m < i$ or $m > j$ and

$$X_{i,j}^- := \prod_{t=i}^{m-1} J(\pi_t^-) \times (Z([\nu^{\alpha_m-l} \rho_m], \Lambda_2^m) \times J((- \Lambda_1^m)^-) \times Y_{3,k_m}^m) \times \prod_{t=m+1}^j J(\pi_t^-)$$

if $i \leq m \leq j$. Note that $X_{1,r}|_{P_n} = \mathbb{X}(\pi_1, \dots, \pi_r, m)$ and $X_{1,r}^-|_{P_n} = \mathbb{X}(\pi_1^-, \dots, \pi_r^-, m)$.

Proof of Lemma 13.13. Note that $J(\pi_r)^{(t)} = 0$ for $1 \leq t < n_r$ and hence any nontrivial P_{α_r, n_r} -submodule of $J(\pi_r)|_P$ has highest derivative at least n_r . By repeatedly applying Lemma 13.8 we obtain that every nonzero $P_{\sum_{i=m+1}^r \alpha_i n_i}$ -submodule of

$$X_{m+1,r}|_P = (J(\pi_{m+1}) \times \dots \times J(\pi_r))|_P$$

has highest derivative at least $n_{m+1} + \dots + n_r$. By Lemma 13.7 2) and Lemma 13.9 any nonzero subrepresentation of

$$X_{m,r}|_P = (Z([\nu^{\alpha_m-1} \rho_m], \Lambda_2^m) \times Y_{3,k_m}^m \times X_{m+1,r})|_P$$

has to have highest derivative at least

$$2n_m + n_{m+1} + \dots + n_r.$$

From this we obtain by repeatedly applying Lemma 13.8 that σ has to have highest derivative at least

$$n_1 + \dots + n_{m-1} + 2n_m + n_{m+1} + \dots + n_r$$

which equals $\lambda_1 + n_m$. $\square$

Proof of Lemma 13.14. Again note that $J(\pi_r)^{(t)} = 0$ for $1 \leq t < n_r$ and hence any nonzero P_{α_r, n_r} -submodule of $J(\pi_r)|_P$ has highest derivative at least n_r . A repeated application of Lemma 13.7 2) and Lemma 13.8 yields that σ has highest derivative at least λ_1 .

First we show that if σ has highest derivative λ_1 then we have an embedding

$$\sigma \hookrightarrow \left(X_{1,r}^- \times \prod_{t=1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

We will prove this using induction, so suppose that for some $1 \leq i \leq r$ we have shown that there is an embedding

$$(13.3) \quad \sigma \hookrightarrow \left(X_{1,i} \times X_{i+1,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

We will then prove that we also have an embedding

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,r}^- \times \prod_{t=i}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

Firstly, for all $1 \leq i \leq r$, we will show that

$$(13.4) \quad X_{i,i} \cong X_{i,i}^- \times \nu^{\alpha_i-1} \rho_i.$$

Recall that for all $i \neq m$ we have

$$X_{i,i} = J(\pi_i) = J(\Lambda_1^i) \times J(\Lambda_2^i) \times \dots \times J(\Lambda_{k_i}^i).$$

Let l_i be the length of Λ_1^i and hence $J(\Lambda_1^i) = \nu^{\alpha_i-l_i} \rho_i \times \nu^{\alpha_i-l_i+1} \rho_i \times \dots \times \nu^{\alpha_i-1} \rho_i$. If $k_i = 1$ we are done. If $k_i \geq 2$ note that for $2 \leq j \leq k_i$ we have that $\nu^{\alpha_i-1} \rho_i$ is unlinked with all factors of Λ_j^i (since, if $k_i \geq 2$, by assumption the length of Λ_1^i is larger than one). This implies for all $2 \leq j \leq k_i$ that

$$\nu^{\alpha_i-1} \rho_i \times J(\Lambda_j^i) \cong J(\Lambda_j^i) \times \nu^{\alpha_i-1} \rho_i.$$

and hence that

$$X_{i,i} \cong X_{i,i}^- \times \nu^{\alpha_i-1} \rho_i.$$

If i equals m , then

$$X_{m,m} = Z([\nu^{\alpha_m-1} \rho_m], \Lambda_2^m) \times J(-\Lambda_1^m) \times Y_{3,k_m}^m$$

and since $\nu^{\alpha_m-1} \rho_m$ is unlinked with all factors of Y_{3,k_m}^m we analogously see that

$$\begin{aligned} J(-\Lambda_1^m) \times Y_{3,k_m}^m &\cong J((- \Lambda_1^m)^-) \times \nu^{\alpha_m-1} \rho_m \times Y_{3,k_m}^m \\ &\cong J((- \Lambda_1^m)^-) \times Y_{3,k_m}^m \times \nu^{\alpha_m-1} \rho_m \end{aligned}$$

and hence

$$X_{m,m} \cong X_{m,m}^- \times \nu^{\alpha_m-1} \rho_m.$$

By putting together Equations (13.3) and (13.4) we obtain an embedding

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,i}^- \times \nu^{\alpha_i-1} \rho_i \times X_{i+1,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

Suppose now that we have shown that there is an embedding

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,j}^- \times \nu^{\alpha_i-1} \rho_i \times X_{j+1,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}$$

for some $i \leq j \leq r-1$. We proceed via induction on j and show that in this case there is also an embedding

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,j+1}^- \times \nu^{\alpha_i-1} \rho_i \times X_{j+2,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

There are four cases.

- If $\nu^{\alpha_i-1} \rho_i \not\cong \nu^{\alpha_{j+1}-1} \rho_{j+1}$, then $\nu^{\alpha_i-1} \rho_i$ is unlinked with all factors in $X_{j+1,j+1}^-$ and we obtain that

$$\nu^{\alpha_i-1} \rho_i \times X_{j+1,j+1}^- \cong X_{j+1,j+1}^- \times \nu^{\alpha_i-1} \rho_i$$

and hence

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,j+1}^- \times \nu^{\alpha_i-1} \rho_i \times X_{j+2,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

- If $\nu^{\alpha_i-1} \rho_i \cong \nu^{\alpha_{j+1}-1} \rho_{j+1}$ and $j+1 \neq m$ we have that $X_{j+1,j+1}^- = J(\pi_{j+1}^-)$ and obtain

$$\nu^{\alpha_i-1} \rho_i \times X_{j+1,j+1}^- \cong J((\Lambda_1^{j+1})^{-,2}) \times \nu^{\alpha_i-1} \rho_i \times \nu^{\alpha_{j+1}-2} \rho_{j+1} \times Y_{2,k_{j+1}}^{j+1}.$$

Since $\nu^{\alpha_i-1} \rho_i \cong \nu^{\alpha_{j+1}-1} \rho_{j+1}$ note that $\nu^{\alpha_i-1} \rho_i \times \nu^{\alpha_{j+1}-2} \rho_{j+1}$ is reducible with the two Jordan-Hölder components $Z([\nu^{\alpha_i-1} \rho_i], [\nu^{\alpha_i-2} \rho_i])$ and $Z([\nu^{\alpha_i-2} \rho_i], \nu^{\alpha_i-1} \rho_i]$. Write Θ for

$$Y_{2,k_{j+1}}^{j+1} \times X_{j+2,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t.$$

Then there is an embedding of σ into

$$\left(X_{1,i-1} \times X_{i,j}^- \times J((\Lambda_1^{j+1})^{-,2}) \times Z([\nu^{\alpha_i-1} \rho_i], [\nu^{\alpha_i-2} \rho_i]) \times \Theta \right) |_{P_n}$$

or into

$$\left(X_{1,i-1} \times X_{i,j}^- \times J((\Lambda_1^{j+1})^{-,2}) \times Z([\nu^{\alpha_i-2} \rho_i], \nu^{\alpha_i-1} \rho_i] \times \Theta \right) |_{P_n}.$$

By Lemma 13.8 any nonzero subrepresentation of $(\prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t) |_P$ has highest derivative at least $n_{i+1} + \dots + n_r$ and then by Lemma 13.7 2) any nonzero subrepresentation of Θ also has highest derivative at least $n_{i+1} + \dots + n_r$. By Lemma 13.9 any nonzero subrepresentation of

$$(Z([\nu^{\alpha_i-1} \rho_i], [\nu^{\alpha_i-2} \rho_i]) \times \Theta) |_P$$

has highest derivative at least $2n_i + n_{i+1} + \dots + n_r$. By applying Lemma 13.7 2) and Lemma 13.8 again we obtain that any nonzero subrepresentation of

$$\left(X_{1,i-1} \times X_{i,j}^- \times J((\Lambda_1^{j+1})^{-,2}) \times Z([\nu^{\alpha_i-1} \rho_i], [\nu^{\alpha_i-2} \rho_i]) \times \Theta \right) |_{P_n}$$

has highest derivative at least $\lambda_1 + n_i$. However, since by assumption σ has highest derivative λ_1 we see that σ can be embedded into

$$\left(X_{1,i-1} \times X_{i,j}^- \times J((\Lambda_1^{j+1})^{-,2}) \times Z([\nu^{\alpha_i-2} \rho_i], \nu^{\alpha_i-1} \rho_i] \times \Theta \right) |_{P_n}.$$

Since

$$Z([\nu^{\alpha_i-2} \rho_i], \nu^{\alpha_i-1} \rho_i] \hookrightarrow \nu^{\alpha_i-2} \rho_i \times \nu^{\alpha_i-1} \rho_i$$

and

$$\nu^{\alpha_i-1} \rho_i \times \prod_{t=2}^{k_{j+1}} J(\Lambda_t^{j+1}) \cong \prod_{t=2}^{k_{j+1}} J(\Lambda_t^{j+1}) \times \nu^{\alpha_i-1} \rho_i$$

we obtain an embedding of σ into

$$\left(X_{1,i-1} \times X_{i,j+1}^- \times \nu^{\alpha_i-1} \rho_i \times X_{j+2,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1} \rho_t \right) |_{P_n}.$$

- If $j+1 = m, l > 2$ and $\nu^{\alpha_i-1}\rho_i \cong \nu^{\alpha_m-1}\rho_m$ recall that

$$X_{m,m}^- = Z([\nu^{\alpha_m-l}\rho_m], \Lambda_2^m) \times J((- \Lambda_1^m)^-) \times Y_{3,k_m}^m.$$

Then $\nu^{\alpha_i-1}\rho_i$ is unlinked with $[\nu^{\alpha_m-l}\rho_m]$ and Λ_2^m and all factors of $J((- \Lambda_1^m)^{-,2})$. Write Ξ_1 for

$$Z([\nu^{\alpha_m-l}\rho_m], \Lambda_2^m) \times J((- \Lambda_1^m)^{-,2})$$

and Ξ_2 for

$$Y_{3,k_m}^m \times X_{j+2,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1}\rho_t.$$

We obtain an embedding of σ into

$$(X_{1,i-1} \times X_{i,j}^- \times \Xi_1 \times \nu^{\alpha_i-1}\rho_i \times \nu^{\alpha_i-2}\rho_i \times \Xi_2) |_{P_n}.$$

By Lemma 13.7 2) and Lemma 13.8 any nonzero subrepresentation of $\Xi_2|_P$ has highest derivative at least $n_{i+1} + \dots + n_r$. Lemma 13.9 implies that any nonzero subrepresentation of

$$(Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i]) \times \Xi_2) |_P$$

has highest derivative at least $2n_i + n_{i+1} + \dots + n_r$ and by Lemma 13.7 2) and Lemma 13.8 we then obtain that any nonzero subrepresentation of

$$(X_{1,i-1} \times X_{i,j}^- \times \Xi_1 \times Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i]) \times \Xi_2) |_{P_n}.$$

has highest derivative at least $\lambda_1 + n_i$. By assumption we then have an embedding of σ into

$$(X_{1,i-1} \times X_{i,j}^- \times \Xi_1 \times Z([\nu^{\alpha_i-2}\rho_i], \nu^{\alpha_i-1}\rho_i) \times \Xi_2) |_{P_n}.$$

Since $Z([\nu^{\alpha_i-2}\rho_i], \nu^{\alpha_i-1}\rho_i) \hookrightarrow \nu^{\alpha_i-2}\rho_i \times \nu^{\alpha_i-1}\rho_i$ and $\nu^{\alpha_i-1}\rho_i \times Y_{3,k_m}^m \cong Y_{3,k_m}^m \times \nu^{\alpha_i-1}\rho_i$ we obtain an embedding

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,m}^- \times \nu^{\alpha_i-1}\rho_i \times X_{m+1,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1}\rho_t \right) |_{P_n}$$

- If $j+1 = m, l = 2$ and $\nu^{\alpha_i-1}\rho_i \cong \nu^{\alpha_m-1}\rho_m$ then

$$X_{m,m}^- = Z([\nu^{\alpha_m-2}\rho_m], \Lambda_2^m) \times Y_{3,k_m}^m$$

and the representation

$$\nu^{\alpha_i-1}\rho_i \times Z([\nu^{\alpha_m-2}\rho_m], \Lambda_2^m)$$

is reducible and by Lemma 13.4 has the two Jordan-Hölder components

$$Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i], \Lambda_2^m)$$

and

$$Z([\nu^{\alpha_i-1}\rho_i, \nu^{\alpha_i-2}\rho_i], \Lambda_2^m).$$

Write Υ for the representation

$$Y_{3,k_m}^m \times X_{m+1,r}^- \times \prod_{t=m}^r \nu^{\alpha_t-1}\rho_t$$

and note that by Lemma 13.7 2) and Lemma 13.8 any nonzero subrepresentation of $\Upsilon|_P$ has highest derivative at least $n_m + \dots + n_r$. Then σ is contained in either

$$(X_{1,i-1} \times X_{i,m-1}^- \times Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i], \Lambda_2^m) \times \Upsilon) |_{P_n}$$

or in

$$(13.5) \quad (X_{1,i-1} \times X_{i,m-1}^- \times Z([\nu^{\alpha_i-1}\rho_i, \nu^{\alpha_i-2}\rho_i], \Lambda_2^m) \times \Upsilon) |_{P_n}.$$

In the first case note that by the theory of Bernstein and Zelevinsky that

$$Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i], \Lambda_2^m) \hookrightarrow Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i]) \times Z(\Lambda_2^m)$$

and hence we have an embedding of σ into

$$(13.6) \quad (X_{1,i-1} \times X_{i,m-1}^- \times Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i]) \times Z(\Lambda_2^m) \times \Upsilon) |_{P_n}.$$

However, by Lemma 13.7 2) and Lemma 13.9 any nonzero subrepresentation of

$$(Z([\nu^{\alpha_i-1}\rho_i], [\nu^{\alpha_i-2}\rho_i]) \times Z(\Lambda_2^m) \times \Upsilon) |_P$$

has highest derivative at least $2n_i + n_{i+1} + \dots + n_r$. By Lemma 13.7 and Lemma 13.8 any nonzero subrepresentation of (13.6) has highest derivative at least $\lambda_1 + n_i$. Hence σ is contained in (13.5), which by Lemma 13.3 implies that σ can be embedded into

$$(X_{1,i-1} \times X_{i,m-1}^- \times Z([\nu^{\alpha_i-2}\rho_i], \Lambda_2^m) \times \nu^{\alpha_i-1}\rho_i \times \Upsilon) |_{P_n}.$$

Since $\nu^{\alpha_i-1}\rho_i$ and all factors of Y_{3,k_m}^m are unlinked we obtain an embedding

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,m}^- \times \nu^{\alpha_i-1}\rho_i \times X_{m+1,r}^- \times \prod_{t=i+1}^r \nu^{\alpha_t-1}\rho_t \right) |_{P_n}.$$

Induction on j shows that

$$\sigma \hookrightarrow \left(X_{1,i-1} \times X_{i,r}^- \times \prod_{t=i}^r \nu^{\alpha_t-1}\rho_t \right) |_{P_n}$$

and via induction on i we obtain that

$$\sigma \hookrightarrow \left(X_{1,r}^- \times \prod_{t=1}^r \nu^{\alpha_t-1}\rho_t \right) |_{P_n}.$$

By Lemma 13.7 1) there is a $\gamma \geq 0$ such that $(\Phi^-)^\gamma(\sigma) \neq 0$ and $(\Phi^-)^\gamma(\sigma)$ can be embedded into

$$(X_{1,r}^-)^{(\gamma)} \times \left(\left[\prod_{t=1}^r \nu^{\alpha_t-1}\rho_t \right] |_P \right).$$

By Lemma 13.8 any nonzero subrepresentation of $(\prod_{t=1}^r \nu^{\alpha_t-1}\rho_t) |_P$ has highest derivative at least λ_1 , which implies that σ has highest derivative at least $\lambda_1 + \gamma$. Since by assumption σ has highest derivative λ_1 we see that γ equals zero and hence

$$\sigma \hookrightarrow X_{1,r}^- \times \left(\left[\prod_{t=1}^r \nu^{\alpha_t-1}\rho_t \right] |_P \right).$$

By taking the λ_1 -derivative and Corollary 4.14 of [BZ] we obtain that

$$\sigma^{(\lambda_1)} \hookrightarrow X_{1,r}^-$$

which finishes the proof. $\square$

Proof of Lemma 13.12. We proceed via induction on l , the length of Λ_1^m . If $l = 1$ this follows from Lemma 13.13. Suppose now that $l > 1$. By Lemma 13.14 we have that $\lambda_1^\sigma \geq \lambda_1$. If $\lambda_1^\sigma > \lambda_1$ we are done. If $\lambda_1^\sigma = \lambda_1$ we can apply the second part of Lemma 13.14 and obtain an embedding

$$\sigma^{(\lambda_1)} \hookrightarrow \mathbb{X}(\pi_1^-, \dots, \pi_r^-, m).$$

Let σ' be an irreducible subrepresentation of $\sigma^{(\lambda_1)} |_P$ and $\lambda^{\sigma'} = (\lambda_1^{\sigma'}, \lambda_2^{\sigma'}, \dots)$ be the highest derivative partition of σ' . Clearly we have an embedding $\sigma' \hookrightarrow \mathbb{X}(\pi_1^-, \dots, \pi_r^-, m)$. Since $\pi_m^- = Z((\Lambda_1^m)^-, \Lambda_2^m, \dots, \Lambda_{k_m}^m)$ and the length of $(\Lambda_1^m)^-$ is $l - 1$ we can apply our induction hypothesis. We obtain that $\lambda_i^{\sigma'} \geq \lambda_{i+1}$ for $1 \leq i \leq l - 1$ and either:

- there exists $1 \leq j < l - 1$ such that $\lambda_j^{\sigma'} > \lambda_{j+1}$,
- or $\lambda_{l-1}^{\sigma'} > \lambda_l + n_m$.

In the first case

$$(((\sigma')^{(\lambda_2)})(\lambda_3)\dots)(\lambda_j^{\sigma'}) \neq 0$$

and hence

$$(((\sigma')^{(\lambda_2)})(\lambda_3)\dots)(\lambda_j^{\sigma'}) \hookrightarrow ((\sigma^{(\lambda_1)})(\lambda_2)\dots)(\lambda_j^{\sigma'}) \neq 0.$$

If $\lambda_j^{\sigma'} = \lambda_{j+1}$ for $1 \leq j < l-1$ we have

$$(((\sigma')^{(\lambda_2)})(\lambda_3)\dots)(\lambda_{l+n_m}) \neq 0$$

and hence

$$(((\sigma')^{(\lambda_2)})(\lambda_3)\dots)(\lambda_{l+n_m}) \hookrightarrow ((\sigma^{(\lambda_1)})(\lambda_2)\dots)(\lambda_{l+n_m}) \neq 0.$$

In both cases we see that $(\lambda_1, \lambda_2, \dots)$ cannot be the highest derivative partition of σ which implies the result. $\square$

We are now almost in a position to prove the Theorem 6.3. The only point missing is the following lemma.

13.3. Proof of Theorem 6.3.

Lemma 13.15. *Let π be a finitely generated representation of G_n . Suppose that each summand in the cosocle of π has highest derivative partition not smaller than or equal to λ . Then $\pi^{\preceq \lambda} = 0$.*

Proof. Let π' be the image of $\bigoplus_{\lambda' \preceq \lambda} W'_\lambda \otimes \text{Hom}(W'_\lambda, \pi) \rightarrow \pi$, i.e. $\pi^{\preceq \lambda} = \pi/\pi'$. If $\pi^{\preceq \lambda}$ is not zero it has an irreducible quotient σ , i.e. we have a surjective map $\pi^{\preceq \lambda} \twoheadrightarrow \sigma$. However, then clearly σ also appears as a quotient of π and hence by assumption has highest derivative partition λ_σ not smaller than or equal to λ . We have a nonzero map $W'_{\lambda_\sigma} \rightarrow \sigma$ which by the projectivity of W'_{λ_σ} gives rise to a nonzero map $W'_{\lambda_\sigma} \rightarrow \pi$, whose image is not contained in π' . This yields a contradiction. $\square$

Proof of Theorem 6.3. The statement we want to prove is that for $1 \leq t \leq r$ let Δ_t be segments such that Δ_i does not precede Δ_j for any $i > j$. Moreover, if for some $1 \leq i, j \leq r$ we have that $\Delta_i \supset \Delta_j$ we assume that $i > j$. Let λ be the highest derivative partition of $Z(\Delta_1, \dots, \Delta_r)$. Then the surjection

$$I(\Delta_1) \times \dots \times I(\Delta_r) \twoheadrightarrow Z(\Delta_1) \times \dots \times Z(\Delta_r)$$

induces an isomorphism

$$(I(\Delta_1) \times \dots \times I(\Delta_r))^{\preceq \lambda} \cong Z(\Delta_1) \times \dots \times Z(\Delta_r).$$

Let Ω be the kernel of the map

$$I(\Delta_1) \times \dots \times I(\Delta_r) \twoheadrightarrow Z(\Delta_1) \times \dots \times Z(\Delta_r).$$

Then, since the functor $V \mapsto V^{\preceq \lambda}$ is right exact, it is enough to prove that $\Omega^{\preceq \lambda} = 0$. By the above Lemma this would follow if any summand of the cosocle of Ω has highest derivative partition not smaller than or equal to λ . By taking contragredients it is hence enough to show that if ω is an irreducible subrepresentation of $\tilde{\Omega}$, then ω has highest derivative partition not smaller than or equal to λ . Now any irreducible subrepresentation of $\tilde{\Omega}$ is contained in a representation of the form $\pi_1 \times \dots \times \pi_r$ where π_i are irreducible representations with supercuspidal supports $\tilde{\Delta}_i$ and who are not all isomorphic to $Z(\tilde{\Delta}_i)$. The result then follows from Proposition 13.11. $\square$

REFERENCES

- [BR] J. Bernstein, *Representations of p-adic groups*, Harvard University, 1992. Lectures by Joseph Bernstein; written by Karl Rumelhart.
- [BZ] J. Bernstein and A. Zelevinsky, *Induced representations of reductive p-adic groups. I*, Ann. Sci. Éc. Norm. Sup. **4** 10 (1977), no. 4, 441–472.
- [Ch] K. Y. Chan, *On the product functor on inner forms of the general linear group over a non-archimedean field*, Transformation Groups (2024).
- [GGS] R. Gomez, D. Gourevitch, and S. Sahi, *Generalized and degenerate Whittaker models*, Comp. Math. **153** (2017), no. 2, 223–256.

- [Go] D. Gourevitch, *Generalized Whittaker and Zelevinsky models for the general linear group*, appendix to E. Lapid and O. Offen, *Explicit Plancherel formula for the space of symplectic forms*, IMRN **18** (2022), 14255–14294.
- [GS] D. Gourevitch and S. Sahi, *Annihilator varieties, adduced representations, Whittaker functionals, and rank for unitary representations of $GL(n)$* , Selecta Math. **19** (2013), no. 1, 141–172.
- [He] Helm, D. *The Bernstein center of the category of smooth $W(k)[GL_n(F)]$ -modules*, Forum Math. Sigma **4** (2016), e11.
- [He2] Helm, D. *Whittaker modules and the integral Bernstein center for GL_n* , Duke Math. J. **165** (2016), no. 9, 1597–1628.
- [MS] A. Mínguez and V. Sécherre, *Représentations lisses modulo ℓ de $GL_m(D)$* , Duke Math. J. **163** (2014), no. 4, 795–887.
- [MW] C. Moeglin and J. L. Waldspurger, *Modeles de Whittaker degeneres pour des groupes p -adiques*, Math. Z. **196** (1987), no. 3, 427–452.
- [Ze] A. Zelevinsky, *Induced representations of reductive p -adic groups. II. On irreducible representations of GL_n* , Ann. Sci. Éc. Norm. Sup. **4** 13 (1980), no. 2, 165–210.

JOHANNES GIRSCH, DEPARTMENT OF MATHEMATICS, IMPERIAL COLLEGE, LONDON, SW7 2AZ, UNITED KINGDOM.
Email address: johannes.girsch@live.de

DAVID HELM, DEPARTMENT OF MATHEMATICS, IMPERIAL COLLEGE, LONDON, SW7 2AZ, UNITED KINGDOM.
Email address: dhelm@imperial.ac.uk